WHU – Otto Beisheim School of Management

Working Paper

Feedback Trading: Strategies during day and night with global interconnectedness*

Alex Kusen and Markus Rudolf†

October 30, 2018

Abstract

Feedback trading strategies have gained much popularity among researchers in the last decades and are used to illustrate how new information based on returns is reflected in the markets. This paper extends previous studies by decomposing the overall return premium and introducing the global feedback trading model. The global feedback trading model assumes an interconnectedness between multiple countries and captures spillovers. Empirical results illustrate two important findings. First, feedback trading strategies differ across markets when distinguishing between day and night returns. Second, evidence for changing feedback trading is provided by examining the interaction of specific markets.

JEL classification:

C58, G02, G10, G11, G12, G14, G15, G17

*The authors would like to thank the seminar participants of the 2017 Paris Financial Management Conference for useful suggestions and comments that have materially improved the paper.

†The corresponding authors work at WHU – Otto Beisheim School of Management, Allianz Endowed Chair of Finance, Burgplatz 2, 56179 Vallendar, Germany.

Electronic copy available at: https://ssrn.com/abstract=3275317

1 Introduction

As one of the first, Beja and Goldman (1980) consider dynamics and trends of investors in a behavioral model setting because trading strategies account as triggers for price movements while market conditions change over time. Feedback trading strategies have gained much popularity among researchers in the last decades and are used to illustrate how new information based on returns is reflected in the markets. This behavioral approach is in contrast to Fama (1970) with his notion of an efficient capital market by stating that security prices fully reveal all available information.

The literature on feedback trading is extensive and its implied effects have still been the subject of research in recent years. Cutler et al. (1990) use a model with feedback traders to account for possible links between volatility and serial correlation. They confirm that fluctuations in speculative prices cannot be predicted under the efficient market hypothesis as demonstrated at least with stock market crashes. Price determination relies on the existence of heterogeneous investors – e.g. informed traders, liquidity traders, fundamental traders, and naive (noise) traders. Thereby, noise investors base their portfolio decisions on the serial correlation properties of stock returns by discovering trends in past stock prices. The so-called feedback traders pursue their own strategy, given explicit market conditions. Positive feedback traders actively buy (sell) stocks in a rising (falling) market while negative feedback traders adhere to a "buy low and sell high" investment strategy.

Stock return autocorrelation increases in the presence of feedback traders. The more feedback traders are active the higher is the autocorrelation due to a greater effect on prices. Hence, the existence of feedback traders influences the partial predictability of aggregate stock returns. Literature and experimental findings support the existence of positive feedback traders while empirical evidence is divided with respect to different stock markets and changing return behavior.

We offer an informative framework for exploring feedback trading from two new angles. First, we decompose the overall return premium into day and night returns. Trading activities during opening hours are based on less noise because prices are more efficient and more information is revealed. Market closures, on the other hand, affect the mean and standard deviation of returns due to risk compensations when trading overnight. Second, we assume interconnectedness between multiple countries and introduce a global feedback trading model. Dynamic effects with spillovers shed light on investors' strategy and provide justifications for price fluctuations.

Electronic copy available at: https://ssrn.com/abstract=3275317

The structure of our research paper is as follows. The foundation of feedback trading and previous empirical work is discussed in section 2. Section 3 states the methodology and extends the feedback trading model. In section 4, analyzed data concerning different market indices and descriptive statistics are provided in addition to developed hypotheses. The main part of this paper consists of the analysis by separating day from night returns in the presence of global interconnectedness in section 5. Furthermore, a robustness check and an outlook are presented in this section. Section 6 concludes.

2 Literature Review

The behavior of different types of investors is a popular field of research in financial studies, especially the study of trading strategies. Several hypotheses and models try to explain the theoretical foundation of strategies, where the distribution of returns depends on current and past performance (see Campbell and Kyle (1993), Conrad et al. (1991), Farmer and Joshi (2002) Jegadeesh and Titman (1993), Kwon and Kish (2002), Long et al. (1989), and Shefrin and Statman (1985) among others). Unanticipated changes provide new information for market participants with corresponding stock price changes in both up and down direction. Noise in the market induces more liquidity and simultaneously more risk, expressed through higher volatility. Shiller et al. (1984) and Shiller (1990) state that fundamentals, i.e. dividends, earnings, and debt, cannot explain asset price volatility. Returns are rather predictable because stock prices do not follow a random walk in irrational markets due to investors' overreactions. This view is confirmed by De Long et al. (1990) who show that prices do not correspond to intrinsic values. Noise trading leads to large price deviations from fundamental values. Thereby, informed traders cannot fully offset the impact of noise traders in a high-risk environment because risk also rises especially during short time horizons.

Different forms of noise trading exist. A destabilizing effect occurs when noise trading strategies correlate in the market. This is the case when market participants conclude relevant information from analyzing same indicators, thus ending up with highly correlated trades and influencing price fluctuations. One of the most common destabilizing forms of noise trading is feedback trading. Positive feedback traders actively buy (sell) stocks in a rising (falling) market while negative feedback traders adhere to a "buy low and sell high" investment strategy. The feedback trading model, originally introduced by Sentana and Wadhwani (1992), allows for changing trading rules in response to market conditions and

Electronic copy available at: https://ssrn.com/abstract=3275317

investors' past experiences.[1] They find a positive feedback trading behavior. Stock prices overreact to news and exhibit excessive volatility. In line with Koutmos (1997), periods with low volatility induce positively autocorrelated returns while autocorrelations turn negative during periods of high volatility. Thereby, the magnitude of autocorrelation is driven by the level of volatility. The inverse relationship between autocorrelation and volatility is mainly due to the interaction between rational expected utility maximizers and trend chasing traders. As a consequence, investors perform a forced liquidation (depyramiding) by withdrawing unprofitable positions with the aim to minimize losses. More feedback trading is expected because most investors insure high-performing assets in their portfolios. The existence of autocorrelation accounts in most studies for the presence of feedback trading. However, the complexity of return autocorrelation is not as elementary as generally assumed according to Conrad et al. (1991) and LeBaron (1992) inversely relates serial return correlations to conditional volatilities by reporting significant nonlinear first moment dependencies.

Overall, strong evidence is found in previous research that feedback trading exists in bond markets (Dean and Faff (2008)), emerging and mature stock markets (Bohl and Siklos (2008)), options markets (Tavakkol (2000)), exchange-traded fund markets (Charteris et al. (2014)), index futures markets (Antoniou et al. (2005)), mutual funds markets (Hsieh et al. (2011)), commodity markets (Chau et al. (2015)), and real estate markets (Koulakiotis and Kiohos (2016)). The intensity is driven by institutional investors (Lakonishok et al. (1992) and Spiegel and Subrahmanyam (1995)), arbitrage opportunities (Chau et al. (2015)), margin requirements (Watanabe (2002)), investor's sentiment (Chau et al. (2011) and Kurov (2008)), cost of credit (Antoniou and Koutmos (2014)), stop-loss orders (Koutmos (2014)), and short sale constraints (Bohl et al. (2013)).

Nevertheless, the feedback trading behavior is not considered when decomposing the overall return premium into day and night returns. Multiple studies confirm that market closures affect investor's trading decisions and return patterns (e.g. Oldfield and Rogalski (1979), Wood et al. (1985), Stoll and Whaley (1990), Madhavan et al. (1997), and Hong and Wang (2000)). In general, the trading activity and volatility decrease at night while stock returns increase. Longstaff (1995) justifies that investors receive a higher reward overnight due to less liquidity. This finding is robust across different market structures and asset types according to Cooper et al. (2008). Furthermore, the information at opening hours is not equivalent to the prior day closure because firms strategically time the release of

[1] The model is based on Cutler et al. (1990) who realize that dynamic price movements arise from heterogeneous investors.

Electronic copy available at: https://ssrn.com/abstract=3275317

information.[2]

Motivated by the fact that feedback trading dynamics are more versatile than originally assumed, we seek to examine the existence of feedback trading during daytime and overnight periods in a multi market setting. Based on the forgoing discussion, we expect less feedback trading during the night because of higher risk, less trading activity, and lower market quality.

3 Methodology

An appropriate measure for volatility effects on the serial correlation of returns is essential in order to answer the proposed research questions. Improvements have been made throughout the years and provide justified approaches to explain potential departures from the efficient market hypothesis.

3.1 Feedback Trading Model

In simple terms, several types of investors carry different implications for the autocorrelation pattern of stock returns. The feedback trading model, as historically adopted by Sentana and Wadhwani (1992), is based on the existence of two types of investors. Thereby, this approach measures investors' behavior due to stock price changes given expectations over a certain period of time. In order to be accurate while using this technique, we consider three heterogeneous groups and develop a dynamic feedback trading model using the following procedure. The advantage of a dynamic interaction between countries is given by estimating the dynamic conditional correlation at each point in time.

Rational utility maximizers (smart money or rational speculators) with an arbitrary von Neumann-Morgenstern utility function hold a fraction of shares of the market portfolio given by:

$$Y_{1,t} = \frac{E_{t-1}(R_t) - r_f}{\theta \sigma_t^2} \qquad \theta > 0, \tag{1}$$

where $Y_{1,t}$ is the fraction of shares held by rational speculators at time t, R_t the ex-post stock return at t, and E_{t-1} the expectation as of time $t-1$. r_f is the risk-free rate of return, σ_t^2 the conditional variance (risk) at t, and θ the coefficient of risk aversion. The required risk premium is defined by the product $\theta\sigma_t^2$ and takes into account that individuals are risk-averse as implied under the expected utility theory.[3]

[2] Firms disclose negative news rather overnight and benefit from a lack of available attention.

[3] If all investors have the same demand function given by eq. (1), the market equilibrium yields the ICAPM as stated by Merton (1973).

Electronic copy available at: https://ssrn.com/abstract=3275317

The second group includes trend chasing investors. Positive feedback traders actively buy (sell) stocks in a rising (falling) market while negative feedback traders adhere to a "buy low and sell high" investment strategy. The feedback function assumes that demand is simply proportional to past return changes of the asset and affected by sentiment in an additive way:

$$Y_{2,t} = \gamma(R_{t-1} - r_f) + \beta B_t \qquad \gamma > 0, \tag{2}$$

where $Y_{2,t}$ is the fraction of shares demanded by feedback traders at time t. γ is the basic feedback parameter and discriminates between two types of feedback traders: $\gamma < 0$ implies a negative feedback behavior while $\gamma > 0$ refers to the case of positive feedback trading. B_t is a dummy variable taking the value one or zero for optimistic and pessimistic investors' sentiment, respectively, at t. The coefficient β measures the sensitivity of feedback traders to the sentiment state.

The third group consists of investors who base their decisions on fundamental values regarding future returns. They increase their demand for shares when fundamental values are absolutely higher in contrast to prices. The underlying demand function for such investors can be expressed as:

$$Y_{3,t} = \delta(f_{t-1} - P_{t-1}) \qquad \delta > 0, \tag{3}$$

where $Y_{3,t}$ is the fraction of shares held by fundamental traders at time t. f_{t-1} and P_{t-1} are the fundamental value and the price, respectively, at time $t-1$. The parameter δ measures the degree of investment.

Given the assumption of market clearing and that all shares must be held at any period, it follows from eq. (1) - (3):

$$Y_{1,t} + Y_{2,t} + Y_{3,t} = 1 \tag{4}$$

or alternatively

$$\frac{E_{t-1}(R_t) - r_f}{\theta\sigma_t^2} + \gamma(R_{t-1} - r_f) + \beta B_t + \delta(f_{t-1} - P_{t-1}) = 1. \tag{5}$$

A regression equation with a stochastic error term can be achieved by setting $R_t - r_f = r_t$ and $E_{t-1}(R_t) - r_f = r_t + \varepsilon_t$:

$$r_t = \theta\sigma_t^2 - \gamma\theta\sigma_t^2 r_{t-1} - \beta\theta\sigma_t^2 B_t - \delta\theta\sigma_t^2(f_{t-1} - P_{t-1}) + \varepsilon_t. \tag{6}$$

The term $-\gamma\theta\sigma_t^2 r_{t-1}$ shows that positive (negative) feedback trading induces negative (positive) first-order autocorrelation in returns and the intensity is driven by σ_t^2. The regression equation can be simplified into the following form:

Electronic copy available at: https://ssrn.com/abstract=3275317

$$r_t = b_0 + b_1\sigma_t^2 + b_2\sigma_t^2 r_{t-1} + b_3\sigma_t^2 B_t + b_4\sigma_t^2 f_{t-1}^* + \varepsilon_t \tag{7}$$

with $b_0 = 0$, $b_1 = \theta$, $b_2 = -\gamma\theta$, $b_3 = -\beta\theta$, $b_4 = -\delta\theta$, and $f_{t-1}^* = f_{t-1} - P_{t-1}$. The coefficient b_1 illustrates the presence of rational speculators and should be positive according to risk aversion under the expected utility theory. b_2 should have a negative (positive) sign when the majority of traders in the market consists of positive (negative) feedback traders. Thereby, high volatility impacts the magnitude of negative autocorrelation. Cyclical market conditions influence investors' sentiment. The coefficient b_3 can either be positive or negative according to significant volatility movements. Simultaneously, a high degree of volatility impacts the presence of fundamental traders because prices move away from fundamental values. The sign of b_4 can be expected to turn in both directions. It is worth mentioning that volatility movements are intensified through the interaction of the three investor types with changing trading behavior. Unlike a statical consideration, an increased volatility from feedback behavior shifts the required rate of return for rational speculators and impacts fundamental values.

Normality of returns' distribution is oftentimes rejected for stock indices.[4] An appropriate method to overcome this issue is the introduction of a volatility term into the mean equation. The conditional variance of the error term is an essential component in eq. (7).

This paper uses the exponential generalized autoregressive conditional heteroscedasticity (EGARCH) model from Nelson (1991) in order to model the conditional variance for a single country by:

$$ln(\sigma_t^2) = \alpha_0 + \sum_{l=1}^{p} \gamma_l ln\sigma_{t-l}^2 + \sum_{l=1}^{q} \alpha_l[|z_{t-l}| - E|z_{t-l}|] + \sum_{l=1}^{q} \xi_l z_{t-l} \tag{8}$$

with $z_{t-l} = \epsilon_{t-l}/\sigma_{t-l}$ as the standardized residual at time $t-l$ and l as the number of lags. $ln(\cdot)$ is the natural logarithm. The main advantage is that asymmetric effects are included between positive and negative returns. Campbell and Hentschel (1992) argue that positive shocks in returns lead to lower volatility than negative shocks given the same magnitude. Furthermore, the EGARCH specification does not require any parameter restrictions. Thereby, this approach is robust against volatility changes and long-run memory effects arising from time-varying clustering in financial time series.

Standardized residuals are rather leptokurtic than normally distributed. Distributions with flatter tails induce more reliable tests in comparison to standard t-tests. Therefore, the computation is based on the Generalized Error Distribution (GED) with the density function:

[4] See Officer (1972), Hagerman (1978), Peiró (1994), and Longin (2005) for more discussion.

Electronic copy available at: https://ssrn.com/abstract=3275317

$$f(\mu_t, \sigma_t^2, \nu) = \frac{\nu}{2}[\Gamma(3/\nu)]^{\frac{1}{2}}[\Gamma(1/\nu)]^{-\frac{3}{2}}(1/\sigma_t)exp\{-[\Gamma(3/\nu)/\Gamma(1/\nu)]^{\frac{\nu}{2}}|z_t|^{\nu}\}, \quad (9)$$

where μ_t is the conditional mean, σ_t^2 the conditional variance, and $\Gamma(\cdot)$ the gamma function. ν is an endogenously estimated scale parameter and depicts the degrees of freedom under GED. The GED yields the normal distribution, the Laplace distribution[5], and converges pointwise to a uniform density for $\nu = 2$, $\nu = 1$ and $\nu \to \infty$ degrees of freedom, respectively.

The maximization of the log-likelihood function with given values of ε_t and σ_t^2 over the sample period induces the fixed parameters in eq. (1) - (9) and can be expressed as:

$$\mathcal{L}(\theta) = \sum_{t=1}^{T} logf(\mu_{t-l}, \sigma_{t-l}^2, \nu). \quad (10)$$

The estimation can be interpreted as a quasi-maximum likelihood method as proposed by White (1982) because error terms are nonnormally distributed.[6] The numerical maximization technique follows either the Broyden-Fletcher-Goldfarb-Shanno or the Berndt-Hall-Hall-Hausman algorithm to avoid nonconcave regions when maximizing the likelihood function. Berndt et al. (1974) justify their approach by a nonlinear log-likelihood function in the parameters.

3.2 Global Feedback Trading Model

In order to estimate the dynamic effects between multiple countries, a multivariate generalized autoregressive conditional heteroscedasticity (MGARCH) model is used. Following Bollerslev et al. (1988), the MGARCH model is a multivariate extension of the GARCH(p,q) model with m regression equations. Conditional variances and covariances of the error terms are dynamic and follow an autoregressive moving average structure. The general MGARCH model is given by:

$$\begin{aligned} \boldsymbol{r}_t &= \boldsymbol{A}\boldsymbol{x}_t + \boldsymbol{\epsilon}_t \\ \boldsymbol{\epsilon}_t &= \boldsymbol{H}_t^{1/2}\boldsymbol{\nu}_t, \end{aligned} \quad (11)$$

where $\boldsymbol{r}_t$ is an $m \times 1$ vector of dependent variables, $\boldsymbol{A}$ an $m \times k$ matrix of parameters, $\boldsymbol{x}_t$ a $k \times 1$ vector of independent variables, $\boldsymbol{\epsilon}_t$ an $m \times 1$ vector of residuals, $\boldsymbol{H}_t^{1/2}$ the Cholesky factor of the time-varying conditional covariance matrix $\boldsymbol{H}_t$, and $\boldsymbol{\nu}_t$ an $m \times 1$ vector with zero-mean, unit variance, and independent identically distributed (iid) innovations at time t for m countries. Thereby, the trade-off be-

[5] The Laplace distribution is often called the double exponential distribution.

[6] Bollerslev and Wooldridge (1992) identify that standard errors are robust under given estimation procedure.

Electronic copy available at: https://ssrn.com/abstract=3275317

tween parsimony and flexibility has to be considered when applying a specific parameterization due to existing comovements of returns. A wide range of parameterizations exists in the literature.[7] Engle (2002) introduces a MGARCH model with dynamic conditional correlation (DCC). The diagonal elements in the matrix $\boldsymbol{H}_t$ represent univariate GARCH models and the off-diagonal elements are modeled as nonlinear functions of the diagonal terms:

$$h_{ij,t} = \rho_{ij,t}\sqrt{h_{ii,t}h_{jj,t}}, \tag{12}$$

where $h_{ii,t}$ and $h_{jj,t}$ follow an univariate and $\rho_{ij,t}$ a dynamic process for the (i,j)th element. The DCC MGARCH model is given by:

$$\begin{aligned}
\boldsymbol{r}_t &= \boldsymbol{A}\boldsymbol{x}_t + \boldsymbol{\epsilon}_t \\
\boldsymbol{\epsilon}_t &= \boldsymbol{H}_t^{1/2}\boldsymbol{\nu}_t \\
\boldsymbol{H}_t &= \boldsymbol{D}_t^{1/2}\boldsymbol{R}_t\boldsymbol{D}_t^{1/2} \\
\boldsymbol{R}_t &= diag(\boldsymbol{Q}_t)^{-1/2}\boldsymbol{Q}_t diag(\boldsymbol{Q}_t)^{-1/2} \\
\boldsymbol{Q}_t &= (1-\lambda_1-\lambda_2)\boldsymbol{R} + \lambda_1\tilde{\boldsymbol{\epsilon}}_{t-1}\tilde{\boldsymbol{\epsilon}}_{t-1}^T + \lambda_2\boldsymbol{Q}_{t-1}.
\end{aligned} \tag{13}$$

$\boldsymbol{D}_t$ is a diagonal matrix of conditional variances:

$$\boldsymbol{D}_t = \begin{bmatrix} \sigma_{1,t}^2 & 0 & \dots & 0 \\ 0 & \sigma_{2,t}^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_{m,t}^2 \end{bmatrix} \tag{14}$$

with

$$\sigma_{i,t}^2 = \alpha_{0,i} + \sum_{l=1}^{p_i}\beta_l\sigma_{i,t-l}^2 + \sum_{l=1}^{q_i}\alpha_l\epsilon_{i,t-l}^2. \tag{15}$$

$\boldsymbol{R}_t$ is a symmetric matrix of conditional quasi correlations[8] and can be expressed as:

$$\boldsymbol{R}_t = \begin{bmatrix} 1 & \rho_{12,t} & \dots & \rho_{1m,t} \\ \rho_{12,t} & 1 & \dots & \rho_{2m,t} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{1m,t} & \rho_{2m,t} & \dots & 1 \end{bmatrix}, \tag{16}$$

where $|\rho_{ij,t}| = |\frac{q_{ij,t}}{\sqrt{q_{ii,t}q_{jj,t}}}| \leq 1$ and $q_{ij,t}$ are the underlying elements in $\boldsymbol{Q}_t$. $\tilde{\boldsymbol{\epsilon}}_t$ is an $m \times 1$ vector of standardized residuals calculated by $\boldsymbol{D}_t^{-1/2}\boldsymbol{\epsilon}_t$ at time t with

[7] See Bauwens et al. (2006), Silvennoinen and Teräsvirta (2009), or Caporin and McAleer (2012) for a meaningful distinction between various MGARCH models.

[8] Aielli (2009) shows that $\boldsymbol{R}$ is neither an unconditional mean nor an unconditional correlation matrix of $\boldsymbol{Q}_t$.

Electronic copy available at: https://ssrn.com/abstract=3275317

$\tilde{\boldsymbol{\epsilon}}_t \sim \mathcal{N}(\mathbf{0}, \boldsymbol{R}_t)$. $\boldsymbol{R} = Cov[\tilde{\boldsymbol{\epsilon}}_t \tilde{\boldsymbol{\epsilon}}_t^T] = E[\tilde{\boldsymbol{\epsilon}}_t \tilde{\boldsymbol{\epsilon}}_t^T]$ is the unconditional covariance matrix of $\tilde{\boldsymbol{\epsilon}}_{\boldsymbol{t}}$ and expressed as a weighted average:

$$\boldsymbol{R} = \frac{1}{T} \sum_{t=1}^{T} \tilde{\boldsymbol{\epsilon}}_t \tilde{\boldsymbol{\epsilon}}_t^T. \tag{17}$$

The nonnegative parameters λ_1 and λ_2 control the conditional quasi correlations. A necessary condition for nonnegativity is satisfied by $0 \leq \lambda_1 + \lambda_2 < 1$. In addition, $\boldsymbol{Q}_0$, as the starting value of $\boldsymbol{Q}_t$, has to be positive definitely to guarantee $\boldsymbol{H}_t$ and $\boldsymbol{R}_t$ to be positive definite. If $\boldsymbol{\nu}_t$ follows a multivariate t-distribution with degrees of freedom (df) greater than two, then the conditional log-likelihood function can be written as:

$$\begin{aligned} \mathcal{L}(\theta) = &\sum_{t=1}^{T} log\Gamma\left(\frac{df+m}{2}\right) - log\Gamma\left(\frac{df}{2}\right) - \frac{m}{2} log[(df-2)\pi] \\ &- \frac{1}{2} log[det(\boldsymbol{R}_t)] - log[det(\boldsymbol{D}_t^{1/2})] - \frac{df+m}{2} log\left(1 + \frac{\tilde{\boldsymbol{\epsilon}}_t \boldsymbol{R}_t^{-1} \tilde{\boldsymbol{\epsilon}}_t^T}{df-2}\right). \end{aligned} \tag{18}$$

The global feedback trading model represents an extension of the feedback trading model by using the DCC MGARCH methodology. Thereby, m regression equations of eq. (7) are implemented with respect to different countries. The DCC MGARCH model with m countries yields the following matrix expression for the global feedback trading model:

$$\underbrace{\begin{bmatrix} r_{1,t} & 0 & \dots & 0 \\ 0 & r_{2,t} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & r_{m,t} \end{bmatrix}}_{\boldsymbol{r_t}} = \underbrace{\begin{bmatrix} b_{1,0} & b_{1,1} & b_{1,2} & b_{1,3} & b_{1,4} \\ b_{2,0} & b_{2,1} & b_{2,2} & b_{2,3} & b_{2,4} \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ b_{m,0} & b_{m,1} & b_{m,2} & b_{m,3} & b_{m,4} \end{bmatrix}}_{\boldsymbol{A}} \underbrace{\begin{bmatrix} 1 & 1 & \dots & 1 \\ \sigma_{1,t}^2 & \sigma_{2,t}^2 & \dots & \sigma_{m,t}^2 \\ \sigma_{1,t}^2 r_{1,t-1} & \sigma_{2,t}^2 r_{2,t-1} & \dots & \sigma_{m,t}^2 r_{m,t-1} \\ \sigma_{1,t}^2 B_{1,t} & \sigma_{2,t}^2 B_{2,t} & \dots & \sigma_{m,t}^2 B_{m,t} \\ \sigma_{1,t}^2 f_{1,t-1}^* & \sigma_{2,t}^2 f_{2,t-1}^* & \dots & \sigma_{m,t}^2 f_{m,t-1}^* \end{bmatrix}}_{\boldsymbol{x_t}} \circ \boldsymbol{I}_m + \underbrace{\begin{bmatrix} \epsilon_{1,t} & 0 & \dots & 0 \\ 0 & \epsilon_{2,t} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \epsilon_{m,t} \end{bmatrix}}_{\boldsymbol{\epsilon_t}}. \tag{19}$$

An implementation of all matrices in eq. (13) would exceed the scope of this paper and is therefore not presented here.[9]

[9] It is obvious that the global feedback trading model follows the same structure as under eq. (13) by assuming $\boldsymbol{r}_t$ as the dependent variable matrix, m as the number of countries, and k as the number of different parameters of independent variables for each country with $k \times m$ parameters in total. To be precise, the $m \times m$ matrix $\boldsymbol{r_t}$, the $m \times 5$ matrix $\boldsymbol{A}$, the $5 \times m$ matrix $\boldsymbol{x}_t$, and the $m \times m$ matrix $\boldsymbol{\epsilon}_t$ are provided in eq. (19). Note that the matrix $\boldsymbol{x}_t$ is defined as the Hadamard product between independent variables and the identity matrix $\boldsymbol{I}_m \in \mathbb{R}^{m \times m}$.

Electronic copy available at: https://ssrn.com/abstract=3275317

4 Data

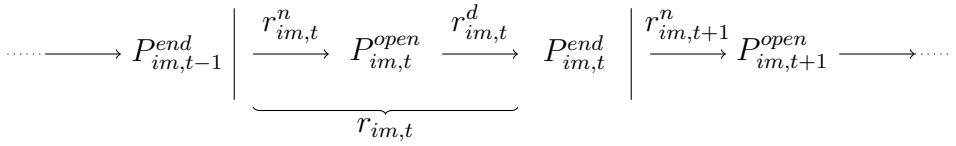
4.1 Sample and Variable Construction

We decompose the overall return premium and compute returns for full trading days (daily returns), trading periods (day returns) and nontrading periods (night returns) as provided in eq. (20). Returns are calculated continuously compounded by:

$$\begin{aligned} r_{im,t} &= ln\left(\frac{P^{end}_{im,t}}{P^{end}_{im,t-1}}\right) * 100 \\ r^{d}_{im,t} &= ln\left(\frac{P^{end}_{im,t}}{P^{open}_{im,t}}\right) * 100 \\ r^{n}_{im,t} &= ln\left(\frac{P^{open}_{im,t}}{P^{end}_{im,t-1}}\right) * 100 \\ r_{im,t} &= r^{d}_{im,t} + r^{n}_{im,t} \end{aligned} \tag{20}$$

with $r_{im,t}$ as the daily return, $r^{d}_{im,t}$ as the day return, and $r^{n}_{im,t}$ as the night return of index i in period t for country m. $P^{end}_{im,t}$ and $P^{open}_{im,t}$ are the closing and opening prices of index i in period t for country m. Figure 1 illustrates the timeline of all three return types according to the definitions in eq. (20).

Figure 1: Return timeline



Note: Continuously compounded returns are presented in this figure. Thereby, $P^{end}_{im,t-1}$ and $P^{end}_{im,t}$ are the closing prices of index i for country m in period $t-1$ and t, respectively. The contrary opening prices are specified as $P^{open}_{im,t}$ and $P^{open}_{im,t+1}$ at time t and $t+1$. $r_{im,t}$ is the daily return, $r^{d}_{im,t}$ the day return, and $r^{n}_{im,t}$ the night return of index i in period t for country m.

Daily observations are used in this paper for stock price indices of five major stock exchanges. The observed group[10] consists of Germany, the UK, the US, Japan, and Hong Kong. We examine individual and mutual interactions of heterogeneous investors and how their actions drive stock prices in these countries. In particular, we use the following indices: DAX 30 (Germany), FTSE 100 (UK), S&P 500 (US), Nikkei 500 (Japan), and Hang Seng Index (Hong Kong). The composition of stocks varies in each index, but still covers at least 80% of the free float-adjusted market capitalization in a specific country. Daily opening and

[10] All presented markets are classified as advanced economies according to the International Monetary Fund.

Electronic copy available at: https://ssrn.com/abstract=3275317

closing prices were collected from Thomson Reuters Datastream. We employ a long estimation window because the variance of the sampling error of the parameters converges to zero. In our case, the sample period covers 2165 trading days and spans from June 27, 2007, to March 7, 2017.[11] Returns are calculated as logarithmic first differences and expressed in percentage terms as shown in eq. (20).

Daily dividend yield and price earnings ratio (P/E) data are taken from Datastream for the same time frame. We calculate dividend payments, D_t, and earnings, E_t, from mentioned fundamental value indicators in order to construct reasonable proxies for financial fundamentals for each period t.[12] We implement two different measures for fundamental values to assure robustness of the results. Proxies for financial fundamentals are defined as follows:

$$f_{1,t} = \frac{1}{d-g} D_t$$
$$f_{2,t} = eE_t,$$

where $1/(d-g)$ is a constant multiple with d as the required rate of return and g as the growth rate according to the constant growth model of Gordon (1959) and implemented by Koutmos (2012). To be in line with previous research and justified by Cutler et al. (1990), we use the inverse of the dividend yield as an equivalent constant multiple. e depicts the long-term average earnings multiple as used by Fama and French (2002). The average trailing P/E is adopted for the long-term multiple as an alternative to a discounted cash flow analysis.

Investor's sentiment has a direct impact on the demand function. We identify optimistic and pessimistic periods by calculating the put call ratio.[13] An optimistic (pessimistic) period implies that the current sentiment indicator is smaller (greater) than one. The sentiment-driven dummy variable B_t indicates whether a market is in a bullish or bearish condition by taking the value one and zero, respectively, at time t.

[11] The starting date depends on the availability of all further variables in order to guarantee the same time frame. Public holidays and trading called offs are already excluded. Trading days are correspondingly matched to obtain an equal number of observation days for every market index. The loss of observations differs across markets ranging between 8.8% and 12.2% for Japan and Germany, respectively.

[12] A general agreement on an appropriate proxy for the fundamental value is missing despite a large body of academic literature. In addition, it is worth mentioning that earnings and dividends are only to some extent correlated. Additional information can be derived from earnings due to alternative dividend payments, such as repurchases, which are not reflected in the dividend yield.

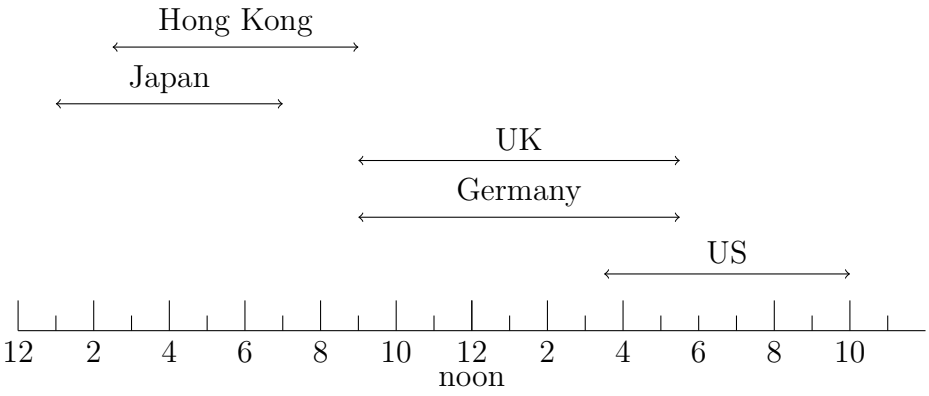
[13] See Baker and Wurgler (2006, 2007) for a detailed analysis of the most renowned sentiment measures.

Electronic copy available at: https://ssrn.com/abstract=3275317

4.2 Descriptive Statistics

Trading hours differ across international markets and stock exchanges operate at the same time, one after the other, or simultaneously. Due to the trading hours overlap illustrated in figure 2, markets cannot be regarded in isolation which can lead to problems with the interpretation of results. The separation of daily returns potentially mislead for the following two reasons. First, it is not possible to distinguish adequately between lagged and contemporaneous effects. Second, information randomly accrues during opening or closing hours of a particular stock exchange and foreign quotes cannot always adjust immediately. Suitable return types are necessary to test for the transmission of information between markets. We eliminate these pitfalls by using independent as well as unbiased day and night returns of stock indices with respect to trading hours. To be precise, we conduct correlations between day and night returns as well as between domestic day (night) and foreign day (night) returns.[14]

Figure 2: Trading hours of stock exchanges



Note: This figure highlights trading hours of major stock exchanges around the world in Central European Time (CET). Lunch break is between 3:30 a.m. and 4:30 a.m. and between 5:00 a.m. and 6:00 a.m. at the Tokyo Stock Exchange and the Stock Exchange of Hong Kong, respectively, in CET.

Pearson correlation coefficients are presented in table 1 for the selected market indices. It can be noticed that most correlations are highly significant.[15] This feature implies both a tight stock market interconnectedness and a strong dependency between return types. Several dependencies show higher correlations be-

[14]The calculations include lagged and leading returns with respect to trading hours and time zones. Day returns of Germany, the UK, and the US are mapped to leading night returns of Japan and Hong Kong. On the contrary, night returns of Japan and Hong Kong are matched with lagged day returns of Germany, the UK, and the US.

[15]High correlations are not due to the fact that we only consider developed countries. Unreported calculations entail similar findings for emerging markets as well. Thereby, we apply three market indices – namely the MSCI Emerging Markets (EM) index, the MSCI EM ex Asia index, and the MSCI World index. The highest correlation is 0.840^{***} (0.785^{***}) between the MSCI World index (DAX) and the MSCI EM ex Asia index.

Electronic copy available at: https://ssrn.com/abstract=3275317

tween day returns compared to night returns. The highest correlation is 0.708*** (0.694***) between Germany and the UK with respect to day (night) returns. Not less interesting is the interaction among return types for a particular country and on a cross-country basis. Day returns of Japan are significantly negative correlated with night returns of Japan (-0.056***) and Hong Kong (-0.403***). In contrast, only day returns of Hong Kong illustrate a negative correlation to night returns of Japan (-0.163***). Different trading hours and time zones have a strong impact on the cross-country comparison. Leading as well as lagged returns have to be considered due to nonsynchronous opening hours. The respective stock exchanges in Germany, the UK, and the US operate while the exchanges in Japan or Hong Kong are closed and vice versa. Consequently, the majority of correlations are higher between those day and night returns since price momentum is driven by day returns outside a particular country. The correlation is 0.137*** between day returns of Germany and night returns of the UK and 0.492*** when considering night returns of Hong Kong. The largest value of 0.656*** is identified for the interaction between day returns of the US and night returns of Hong Kong.

Table 2 provides a first analysis of major market indices in terms of descriptive statistics. Thereby, panel (A) depicts day returns and panel (B) night returns for all treated countries in order to compare on a cross-country and cross-return type basis.[16] The aim is to examine the returns' distribution first before proceeding with the feedback trading investigation. Noteworthy is the difference between day and night returns when taking the mean into consideration. The mean is mostly negative for day returns and positive for night returns. This implies that returns are rather negative during trading hours than during closing periods for the following two reasons.[17] First, market uncertainty after market closures and a reduced possibility of liquidation have an impact on night return patterns. Intuitively, a positive risk premium is earned overnight by taking naturally a riskier side of the trade. Second, institutions have a dominant market position and prefer to trade against the momentum characteristic during day in contrast to individual market participants. As expected for financial time series, all return distributions are skewed and leptokurtic relative to the normal distribution. A goodness-of-fit test is highlighted by the Jarque-Bera test, which matches the skewness and kurtosis to the normal distribution. The values range from 3501.561*** to 18568.940*** and from 441.376*** to 180522.400*** for day and night returns, respectively. It is notable that the test statistics are at least

[16]Figure A1 presents opening prices and day returns, whereas figure A2 highlights closing prices and night returns on page 38f. in the appendix. The corresponding conditional variances of returns are provided in figure A3 on page 40.

[17]Unreported calculations confirm that daily returns are positive on average for every country.

Electronic copy available at: https://ssrn.com/abstract=3275317

Table 1: Correlation matrix for stock market returns

	Day returns						Night returns				
	Germany	UK	US	Japan	Hong Kong		Germany	UK	US	Japan	Hong Kong
(A) Day returns						(C) Day returns					
Germany	1					Germany	.075***	.137***	.142***	.425***	.492***
UK	.708***	1				UK	.396***	.177***	.252***	.392***	.385***
US	.587***	.548***	1			US	.162***	.129***	.052**	.467***	.656***
Japan	.076***	.318***	.149***	1		Japan	.406***	.252***	.331***	-.056***	.403***
Hong Kong	.148***	.289***	.284***	.340***	1	Hong Kong	.217***	.087***	.151***	-.163***	.007
(B) Night returns						(D) Night returns					
Germany	.075***	.396***	.162***	.406***	.217***	Germany	1				
UK	.137***	.177***	.129***	.252***	.087***	UK	.694***	1			
US	.142***	.252***	.052**	.331***	.151***	US	.472***	.504***	1		
Japan	.425***	.392***	.467***	-.056***	-.163***	Japan	.220***	.189***	.042*	1	
Hong Kong	.492***	.385***	.656***	-.403***	.007	Hong Kong	.520***	.439***	.399***	.403***	1

Note: The overall return premium is decomposed into day and night returns. Trading hours differ across international markets. Therefore, panel (A) illustrates the correlation between day returns and (D) the correlation between night returns. (B) presents correlations for the combinations between night and day returns, whereas lagged day returns from Germany, the UK, and the US are mapped to night returns in Japan and Hong Kong. In contrast, (C) shows the correlation matrix for day and night returns with leading night returns of Japan and Hong Kong for day returns in Germany, the UK, and the US. Stars indicate significance as follows: (*) 10%, (**) 5%, (***) 1%.

Electronic copy available at: https://ssrn.com/abstract=3275317

10 times higher for night returns of Germany, the UK, and the US than for the corresponding day returns. Nonnormal distributions are mainly driven through an increased kurtosis overnight hours. Stationarity is confirmed by the Augmented Dickey-Fuller (ADF) test. The existence of a unit root can be rejected for all return series. T-statistics for day and night returns have similar significant negative values between -44.303*** and -59.175*** in panels (A) and (B) with respect to the country. Hence, the process is mean-reverting and stochastic shocks diminish with a nonpermanent effect over time. Serial correlation is present in most market indices as the Breusch-Godfrey (BG) Lagrange multiplier (LM) test for up to two lags confirms, especially when considering night returns. The largest dependency between lagged and present day returns can be observed in Japan (9.989***) while in Germany for night returns (15.835***). The Ljung-Box statistic illustrates intertemporal dependencies in both returns and squared returns for up to 12 lags. Panels (A) and (B) show that autocorrelations of most return series are significantly different from zero – implying a dependent distribution. The test statistics for squared returns are smaller when considering night returns and range from 59.529*** to 462.441***. In contrast, values for day returns lie between 805.946*** and 2130.138***. The presence of autoregressive conditional heteroscedasticity (ARCH) is significant for all day returns at the 1% level by applying Engle's LM with two lags. LM-test values are only significant for Germany, Japan, and Hong Kong when considering night returns. The test statistics are obviously smaller for night returns than for day returns. The largest ARCH effects are detected for Hong Kong with values of 455.072*** and 155.807*** for day and night returns, respectively. Therefore, the sample is consistent with current asset pricing literature in terms of ARCH and volatility clustering.[18]

4.3 Hypotheses

Theoretically, serial correlation in returns can arise from other aspects besides feedback trading. Both nonsynchronous trading and investors' preferences are possible explanations for underlying correlation. However, negative autocorrelation in returns due to nonsynchronous trading of an individual security is dominated by positive cross-correlation.[19] The degree of risk aversion in preferences can be neglected because the estimation does not include any specific utility function or exogenous wealth. Hence, investors adapt their behavior differently to market conditions, past performance, and fundamental values. Cooper et al. (2008) argue that the rate of information flow changes around the clock. This

[18] Correlograms of autocorrelations are provided for different stock markets with respect to the return type in figures A4 and A5 on page 41f. in the appendix.

[19] See Lo and MacKinlay (1990) or Sentana and Wadhwani (1992) for a broader discussion.

Electronic copy available at: https://ssrn.com/abstract=3275317

Table 2: Descriptive statistics of stock market returns

	Germany	UK	US	Japan	Hong Kong
(A) Day returns					
Mean	−0.015	−0.010	0.015	−0.036	−0.071
St.Dev.	1.262	1.266	1.261	1.136	1.157
Skewness	0.049	−0.358	−0.565	−0.613	0.244
Kurtosis	8.940	9.189	12.267	15.303	17.339
Jarque-Bera	3183.741***	3501.561***	7862.034***	13 789.867***	18 568.940***
ADF	−44.403***	−46.872***	−53.223***	−48.656***	−51.385***
BG LM (2)	0.684	2.736	0.302	9.989***	6.313**
Ljung-Box (12)	19.950*	16.896	18.698*	30.381***	18.668*
Ljung-Box^2 (12)	805.946***	2130.138***	1690.445***	1896.442***	1179.227***
LM ARCH (2)	173.425***	274.229***	135.172***	438.834***	455.072***
N	2165	2165	2165	2165	2165
(B) Night returns					
Mean	0.024	−0.007	0.006	0.048	0.075
St.Dev.	1.293	0.957	0.563	0.988	1.357
Skewness	−0.009	1.460	4.837	−0.159	0.309
Kurtosis	22.514	39.532	116.987	5.189	16.903
Jarque-Bera	34 351.019***	121 160.023***	180 522.400***	441.376***	17 471.129***
ADF	−50.078***	−45.419***	−46.845***	−59.175***	−49.614***
BG LM (2)	15.835***	5.697*	9.155**	9.737***	0.450
Ljung-Box (12)	34.588***	41.182***	63.492***	15.305	10.550
Ljung-Box^2 (12)	127.356***	59.529***	102.451***	105.029***	462:441***
LM ARCH (2)	33.726***	2.163	2.282	46.608***	155.807***
N	2165	2165	2165	2165	2165

Note: This table provides the sample statistics for each market using day and night returns from June 27, 2007, to March 7, 2017, with a sample size of 2165 observations. All series are in logarithmic first differences form and expressed as a percentage on a daily basis. Jarque-Bera test matches the skewness and kurtosis to the normal distribution's. ADF is the Augmented Dickey-Fuller test for a unit root and BG LM is the Breusch-Godfrey test of order 2 for detecting serial correlation. The Ljung-Box statistic of order 12 for returns and squared returns is defined as Ljung-Box (12) and Ljung-Box^2 (12), respectively. LM ARCH (2) performs Engle's Lagrange multiplier (LM) test for the presence of autoregressive conditional heteroskedasticity. Stars indicate significance as follows: (*) 10%, (**) 5%, (***) 1%.

Electronic copy available at: https://ssrn.com/abstract=3275317

in turn induces larger price volatility due to increased or decreased demand for particular stocks – away from the predetermined market equilibrium. Comprising these thoughts leads directly to the hypotheses:

H0: Effects of volatility on the serial correlation pattern of returns do not exist in the market. Traders' behavior is irrelevant for the pricing of stocks and no feedback traders can be found in the market.

H1: Effects of volatility on the serial correlation pattern of returns exist in the market. Traders change their behavior according to market conditions and past performance, which is in line with a feedback trading strategy.

Analyzing possible drivers of volatility implies a direct prediction concerning the rejection of the null hypothesis in favor of the alternative hypothesis. In addition, we can investigate further implications if the null hypothesis is rejected. Time differences can impact potentially on returns. Barclay and Hendershott (2003) identify asymmetric patterns. They argue that prices are less efficient and less information is revealed at night than during day due to noise, but a single trade contains more information after hours.[20] Market closures affect the mean and standard deviation of returns because investors need to be compensated for the risk when trading overnight as stated by Blanc et al. (2014). Lou et al. (2015) confirm that a positive risk premium is earned at night and the premium offsets intraday average returns. The limited possibility of liquidation constricts night returns to be higher than day returns. Institutional liquidity providers are less active during the night and liquidity traders mostly dominate the the after-hours session according to Giannetti et al. (2006). These can result in either a similar trading behavior or different strategies across the day.

H2: Feedback trading strategies behave similarly during a full trading day.

H3: Feedback trading strategies differ when taking the distinction between day and night returns into account.

Furthermore, the interconnectedness between multiple countries considers dynamic effects according to market spillovers. Price fluctuations in one country

[20]Not surprisingly, the trading volume decreases and trading costs increase at night due to a lack of liquidity, larger quote spreads, uncertain prices, competition with professional traders, price volatility, computer delays, and biases toward limit orders.

Electronic copy available at: https://ssrn.com/abstract=3275317

can certainly influence investors' strategy in another country with a mutual reaction to investors' behavior in the initiating country. Aretz and Bartram (2015) verify that prices are driven by day returns from abroad. According to Frino and Hill (2000), overseas information releases and contagion effects are the main reasons for spillovers.[21] We apply a joint analysis of stock prices with matched day and night returns across countries to control for spillovers in volatility of the lagged returns from one stock exchange to another. Possible interrelations between countries are examined with the following hypotheses:

H4: Spillover effects across international markets do not change investors' behavior. Feedback trading patterns can be individually estimated for each particular country.

H5: Interaction of specific markets has an influence on feedback trading patterns. Stock prices dependent on business cycles and national economic shocks with cross-country spillover effects.

5 Empirical Results

In this part we investigate our implied hypotheses and present results with respect to market segmentation, time differences and global interconnectedness. First, the serial correlation pattern of returns is analyzed for individual markets. Thereby, we distinguish between day and night returns and examine effects arising after market closures as well as during a trading day. We continue with a joint analysis of stock prices on a multi country foundation with the aim to find spillovers in the volatility of lagged returns. Our study ends with a further outlook after applying sophisticated robustness checks.

5.1 Day and Night

Heterogeneous trading strategies of various market participants can have a significant impact on stock prices. The aim is to answer the question whether feedback trading is evident and if feedback trading strategies differ during day and night. An examination of hypotheses 2 and 3 can only be performed when the null hypothesis is rejected. Proceeding from here, one can seek differences and similarities in serial correlation patterns for different return types. Tables 4

[21] Local traders react to foreign (public) information by observing price movements. Thereby, Becker et al. (1995) show that overseas macroeconomic information releases can not only increase asymmetries between overseas and local markets but also amongst market participants in the local market.

Electronic copy available at: https://ssrn.com/abstract=3275317

and 3 report parameter estimation results for eq. (7) and (8), respectively. In particular, table 4 depicts mean equations' estimates for day and night returns with respect to the applied fundamental value, whereas table 3 presents variance equations' estimates for the same underlying mean equation by applying an EGARCH(1,1) model with GED. The technical reason for this is that conditional variances of stock returns are time-varying and depend asymmetrically on past estimates.

The Akaike information criterion (AIC) discloses low values around 2.744 and 2.645 for day and night returns, respectively. As a direct implication, our applied model with one lag is superior in terms of the trade-off between bias and variance in model construction as already demonstrated in figures A4 and A5. As it can be seen by the sign of the mostly significant parameter ξ_1 in table 3, an asymmetric response of the conditional variance to past residuals is observed for day and night returns. This implies that the conditional variance rises faster for negative residuals than for positive ones.[22] Thereby, the volatility feedback hypothesis serves as an important determinant of asymmetric volatility. Negative shocks and unexpected return changes lead to a higher volatility than positive innovations given an equal magnitude as mentioned by Campbell and Hentschel (1992) amongst others.

According to the endogenously estimated scale parameter ν, a distribution with flatter tails has to be assumed for all return types due to leptokurtic standardized residuals. Degrees of freedom between 1.278^{***} (0.977^{***}) and 1.521^{***} (1.542^{***}) for day (night) returns imply rather a Laplace than a normal distribution. The quasi-maximum likelihood estimation yields values around -2910.867 on average with higher values for day than night returns in absolute terms.[23] Unfortunately, it is not possible to estimate valid results for night return patterns in the US due to a flat likelihood. A missing uphill direction with nonconcave regions illustrates obviously the complexity of the model. Nevertheless, it can be assumed that investors' behavior in the US is similar to all other discussed markets.

The adjusted R^2 varies across markets, return types, and fundamental values. In general, the implied feedback trading model explains on average 2.2% (2.5%) of returns' variance during day (night). Barclay and Hendershott (2003) confirm that a significant price discovery can be achieved after hours even with relatively little trading and noisier prices. On a first glance, the fundamental value f_2 suits best when dealing with day returns and f_1 for night returns on average. Markets with different microstructures exhibit unequal price discovery

[22] The news response functions reinforce an asymmetric pattern for both day and night returns in figure A6 on page 43 in the appendix.

[23] Log-likelihoods are on average -2958.774 (-2850.011) and -2958.073 (-2852.834) for day (night) returns given f_1 and f_2, respectively.

Electronic copy available at: https://ssrn.com/abstract=3275317

processes after earnings and dividend announcements according to Greene and Watts (1996). They find that price responses to earnings announcements are mainly realized during the day. Along with Spiegel and Subrahmanyam (1995), institutional investors have a trading advantage and mainly trade at the open or during periods of unusual demand. Thereby, this group of investors typically prefers nonpaying dividend firms in contrast to individual investors as stated by Jain (2007).[24] For day returns, the highest values are 5.9% and 6% for Germany with respect to f_1 and f_2. In contrast, the smallest explanatory power is derived for Japan based on day returns. Values for night returns differ between 4.9% in Japan and 0.7% in the UK for f_1 and f_2, respectively. However, changes in the explanatory power arise when considering day returns. In Hong Kong, for example, the explanatory power with 2.5% is higher by taking f_1 as the fundamental value than under f_2 with an adjusted R^2 of 2.3%. Therefore, a consideration of only one of both fundamental values is, ceteris paribus, not Pareto efficient for further analysis.

We now turn our attention to traders' behavior by examining the effects of volatility on the serial correlation pattern of returns for the different return types and respective countries. Overall, findings affirm the presence of positive feedback traders in daily stock market prices. The coefficient parameter b_2 is mostly significantly negative with values ranging from -0.174*** to -0.007* in table 4. The null hypothesis can be rejected in favor of the alternative hypothesis because traders change their behavior according to market conditions and past performance by actively buying (selling) stocks in a rising (falling) market. However, strategies of trend chasing investors differ with regards to trading time and allow us to investigate further implications. Achieving this objective involves, on the one hand, a differentiation between return types and, on the other hand, a consideration of the underlying fundamental value.

Fundamental values are instructive and serve as useful benchmarks for investors to measure over- or undervaluation in stock markets. The crucial parameter coefficient b_4 is mostly negative and statistically significant for all countries and return types when using f_1 or f_2 as the proxy. Only a positive value of 0.058** is estimated for day returns in the US, given f_1. Consistent results across markets suggest that findings are robust regarding fundamental traders.

The coefficient for the dummy variable B_t is always significantly positive

[24]First, the relative proportion of news releases is higher during the day than at night as Patell and Wolfson (1982) highlight. Second, Francis et al. (1992) find that disclosures contain less bad news and surprises during day. Therefore, this systematic timing behavior has a direct impact on price changes and the adjustment process. Furthermore, Lamont (1998) argue that earnings contain more information regarding business conditions and are embedded in press releases with many other bits of relevant information according to Patell and Wolfson (1984).

Electronic copy available at: https://ssrn.com/abstract=3275317

Table 3: Variance equations for day and night returns

Coefficient	Germany	UK	US	Japan	Hong Kong
(A) Day returns with f_1 as the fundamental proxy					
α_0	-0.101***	-0.144***	-0.134***	-0.164***	-0.100***
	(0.017)	(0.020)	(0.018)	(0.022)	(0.015)
α_1	0.133***	0.182***	0.162***	0.200***	0.127***
	(0.022)	(0.026)	(0.024)	(0.029)	(0.019)
ξ_1	-0.108***	-0.149***	-0.180***	-0.099***	-0.049***
	(0.015)	(0.017)	(0.020)	(0.018)	(0.012)
γ_1	0.979***	0.967***	0.973***	0.954***	0.989***
	(0.005)	(0.005)	(0.005)	(0.008)	(0.004)
ν	1.337***	1.521***	1.299***	1.285***	1.467***
	(0.052)	(0.071)	(0.053)	(0.031)	(0.066)
Log-likelihood	-3168.477	-3066.954	-2878.993	-2779.925	-2899.521
Adj. R^2	0.059	0.001	0.008	0.000	0.025
AIC	2.938	2.844	2.670	2.578	2.689
(B) Day returns with f_2 as the fundamental proxy					
α_0	-0.102***	-0.150***	-0.151***	-0.167***	-0.096***
	(0.017)	(0.021)	(0.019)	(0.022)	(0.015)
α_1	0.135***	0.190***	0.186***	0.204***	0.122***
	(0.022)	(0.027)	(0.026)	(0.029)	(0.019)
ξ_1	-0.109***	-0.139***	-0.161***	-0.093***	-0.052***
	(0.015)	(0.017)	(0.020)	(0.017)	(0.012)
γ_1	0.978***	0.968***	0.972***	0.954***	0.988***
	(0.005)	(0.006)	(0.005)	(0.008)	(0.004)
ν	1.339***	1.501***	1.278***	1.289***	1.468***
	(0.052)	(0.070)	(0.052)	(0.031)	(0.066)
Log-likelihood	-3168.725	-3062.940	-2880.079	-2780.636	-2897.983
Adj. R^2	0.060	0.013	0.023	0.003	0.023
AIC	2.938	2.840	2.671	2.579	2.688
(C) Night returns with f_1 as the fundamental proxy					
α_0	-0.091***	-0.076***		-0.097***	-0.107***
	(0.014)	(0.014)		(0.020)	(0.014)
α_1	0.129***	0.101***		0.120***	0.142***
	(0.020)	(0.019)		(0.025)	(0.019)
ξ_1	-0.008	-0.020*		-0.016	-0.066***
	(0.014)	(0.012)		(0.014)	(0.012)
γ_1	0.986***	0.990***		0.974***	0.990***
	(0.004)	(0.003)		(0.009)	(0.003)
ν	1.093***	0.977***		1.542***	1.208***
	(0.028)	(0.029)		(0.057)	(0.042)
Log-likelihood	-3124.226	-2313.100		-2898.319	-3064.398
Adj. R^2	0.018	0.008		0.049	0.034
AIC	2.897	2.147		2.688	2.841
(D) Night returns with f_2 as the fundamental proxy					
α_0	-0.086***	-0.070***		-0.101***	-0.107***
	(0.013)	(0.013)		(0.021)	(0.015)
α_1	0.121	0.089***		0.125***	0.142***
	(0.019)	(0.017)		(0.025)	(0.020)
ξ_1	-0.018	-0.026**		-0.018	-0.075***
	(0.014)	(0.012)		(0.015)	(0.013)
γ_1	0.987***	0.991***		0.973***	0.989***
	(0.004)	(0.003)		(0.009)	(0.003)
ν	1.097***	0.990***		1.537***	1.212***
	(0.028)	(0.029)		(0.056)	(0.042)
Log-likelihood	-3127.728	-2317.743		-2897.535	-3068.330
Adj. R^2	0.011	0.007		0.046	0.025
AIC	2.900	2.151		2.687	2.845

Note: This table presents parameter coefficients for the EGARCH(1,1) specification in order to estimate the conditional variance $ln(\sigma_t^2) = \alpha_0 + \gamma_1 ln\sigma_{t-1}^2 + \alpha_1[|z_{t-1}| - E|z_{t-1}|] + \xi_1 z_{t-1}$. The model is estimated simultaneously with eq. (7) with different fundamental proxies (f_1 and f_2) with respect to day and night returns. Stars indicate significance as follows: (*) 10%, (**) 5%, (***) 1%. In parentheses: standard errors.

Electronic copy available at: https://ssrn.com/abstract=3275317

Table 4: Mean equations for day and night returns

Coefficient	Germany	UK	US	Japan	Hong Kong
(A) Day returns with f_1 as the fundamental proxy					
b_0	0.077**	-0.019	0.071***	0.018	-0.050
	(0.039)	(0.026)	(0.023)	(0.027)	(0.030)
b_1	-0.120***	-0.131**	0.099**	-0.268***	0.015
	(0.057)	(0.059)	(0.045)	(0.076)	(0.085)
b_2	0.000	-0.010*	-0.016***	-0.013***	-0.014**
	(0.005)	(0.006)	(0.006)	(0.005)	(0.006)
b_3	0.280***	0.270***	0.119**	0.322***	0.046
	(0.031)	(0.032)	(0.061)	(0.035)	(0.037)
b_4	-0.019	-0.021	0.058**	-0.016**	0.011
	(0.019)	(0.017)	(0.026)	(0.007)	(0.024)
(B) Day returns with f_2 as the fundamental proxy					
b_0	0.093	-0.041	0.020	0.013	-0.063**
	(0.032)	(0.027)	(0.024)	(0.027)	(0.026)
b_1	-0.136***	-0.008	0.041	-0.067**	-0.007
	(0.030)	(0.031)	(0.030)	(0.038)	(0.032)
b_2	0.000	-0.010*	-0.018***	-0.014***	-0.013**
	(0.005)	(0.005)	(0.005)	(0.005)	(0.006)
b_3	0.270***	0.280***	0.154***	0.324***	0.061*
	(0.031)	(0.032)	(0.055)	(0.035)	(0.035)
b_4	-0.011	-0.010***	-0.065	-0.008**	-0.027*
	(0.015)	(0.013)	(0.047)	(0.004)	(0.014)
(C) Night returns with f_1 as the fundamental proxy					
b_0	0.020	-0.030		-0.094*	0.049**
	(0.034)	(0.019)		(0.054)	(0.022)
b_1	-0.167***	-0.243***		-0.060	-0.210***
	(0.054)	(0.050)		(0.108)	(0.060)
b_2	-0.009**	0.003		-0.174***	-0.007*
	(0.004)	(0.006)		(0.016)	(0.004)
b_3	0.123***	0.081***		0.098*	0.067*
	(0.025)	(0.026)		(0.053)	(0.025)
b_4	-0.012**	-0.021***		-0.020**	-0.060***
	(0.017)	(0.016)		(0.010)	(0.017)
(D) Night returns with f_2 as the fundamental proxy					
b_0	0.071**	0.018		-0.106*	0.080
	(0.030)	(0.015)		(0.056)	(0.020)
b_1	-0.031	-0.029		0.214***	0.011
	(0.026)	(0.025)		(0.072)	(0.018)
b_2	-0.006	0.002		-0.174***	-0.006
	(0.004)	(0.007)		(0.016)	(0.004)
b_3	0.116***	0.064**		0.102*	0.056**
	(0.025)	(0.026)		(0.053)	(0.025)
b_4	-0.004	-0.001		-0.012**	-0.025**
	(0.016)	(0.010)		(0.005)	(0.010)

Note: The estimated parameters rely on the model $r_t = b_0 + b_1\sigma_t^2 + b_2\sigma_t^2 r_{t-1} + b_3\sigma_t^2 B_t + b_4\sigma_t^2 f_{t-1}^* + \varepsilon_t$. Parameters are estimated simultaneously with eq. (8) with respect to day and night returns as well as different fundamental proxies (f_1 and f_2). Stars indicate significance as follows: (*) 10%, (**) 5%, (***) 1%. In parentheses: standard errors.

Electronic copy available at: https://ssrn.com/abstract=3275317

except for day returns in Hong Kong with f_1 as the fundamental proxy. The sentiment level has a direct impact on trend chasing investors by providing a useful signal for trading. The simultaneous existence of positive feedback traders illustrates that investors rather buy stocks in a bullish market situation and vice versa. Therefore, this dummy confirms a significant linkage between the sentiment level and feedback trading and affirms the existence of positive feedback behavior. This finding is in line with Kurov (2008), who confirms that the degree of positive feedback trading is intensified by high-sentiment periods.

We conclude from the empirical results that feedback trading strategies differ across markets when taking day and night returns into account. Feedback traders behave similarly in Japan and Hong Kong during a full trading day, whereas Germany and the UK register a strong contrariness. For Japan and Hong Kong, panels (A) and (B) in table 4 denote the presence of significantly negative b_2-values with respect to both fundamental proxies. Similar results are reported for night returns in panels (C) and (D). Of particular interest are the diverse strategies in Germany and the UK. No feedback trading is found to exist in Germany during trading hours because b_2 is insignificant according to f_1 and f_2. Simultaneously, the parameter is significantly negative at least at the 5% level for night returns when f_1 is the fundamental proxy. An inverse relationship is found in the UK. During day, investors follow a positive feedback trading strategy as the significantly negative values of b_2 illustrate. The absence of feedback trading during night hours, as illustrated by insignificant b_2-values, can reflect a change in investors' behavior in this particular country. Furthermore, an important reason is the systematic trading behavior of different investor groups. Usually, institutional investors engage in positive feedback trading while individual investors follow a negative feedback behavior due to limit trade imbalances according to Lee (2016). Expected return reversals reinforce the negative trading pattern in addition. The lack of feedback trading during day in Germany and overnight in the UK can stem from the fact that positive and negative feedback trading strategies offset each other. This evidence supports our hypothesis that trading strategies differ between day and night hours across our sample markets.

It is not surprising that positive feedback trading tends to occur more often for day than night returns. One possible explanation is the obvious decline in trading volume overnight arising from the clientele effect. On the one hand, trading in the after-hours session is the privilege of professional investors due to high risk levels as stated by Barclay and Hendershott (2003). On the other hand, less sophisticated market participants and institutions with a dominant market position prefer to trade during day. Giannetti et al. (2006) show that institutional

Electronic copy available at: https://ssrn.com/abstract=3275317

liquidity providers are less active during night. Hence, a decline in the trading volume diminishes the attractiveness of feedback behavior and the amount of chasing investors reduces.

In the next part of this analysis, we focus on day and night returns of Germany and the UK, respectively, to assess the impact of spillovers by taking market interconnectedness into consideration. The purpose is to explain potential changes in investors' strategies and to address the question whether feedback trading is still absent in these countries when considering the respective return type.

5.2 Global Interconnectedness

Examining the interaction of specific markets is a superior way to shed light on investors' behavior by taking spillover effects into account. We use the global feedback trading model to estimate parameters and present results in tables 5 as well as 6. Thereby, table 5 incorporates combinations of day returns with respect to Germany and table 6 night returns according to the UK.

In general, the necessary condition for nonnegativity is satisfied for all combinations and with regards to both fundamental proxies. Average values for $\lambda_1 + \lambda_2$ are between 0.577 and 0.952 for day and night returns, respectively, given f_2. An absent unit root in the variance, as the second necessary condition, is achieved according to $\alpha_1 + \beta_1 < 1$. All estimated values are significantly smaller and imply that past shocks diminish after periods of time.[25] Quasi correlations differ across country combinations. The value ρ is larger for the interconnectedness between Germany and the UK (or the US) than between Germany and Japan (or Hong Kong) according to day returns. This circumstance is justified by time-delayed opening and closing hours in a particular country. Day returns in Germany, the UK, and the US have an overlap time. Japan and Hong Kong, on the other hand, are ahead in time and act as leaders with leading signals. Same findings hold for night returns in table 6.[26] Not surprisingly, likelihood estimations are larger in the global feedback trading model compared to the feedback trading model for both return types and fundamental values in absolute terms. Values range from -5580.747 (-4878.089) to -6099.705 (-5247.037) for day (night) returns. In contrast to a GED assumption, ν_t follows here a multivariate t-distribution. All

[25] The persistence of volatility can be measured by the concept of half-life, which discloses the number of periods it takes to move halfway towards the unconditional volatility, and is defined as: $halflife = \frac{-ln(2)}{ln(\alpha_1+\beta_1)}$. Volatility has a half-life of about 105 days on average with respect to day and night returns, whereas the mean reversion lasts longer for night returns (156 days) than for day returns (54 days).

[26] Note that night returns of the US are again not considered due to a flat likelihood in the previous feedback trading model.

Electronic copy available at: https://ssrn.com/abstract=3275317

parameters in the model are estimated along with df. This approach is robust for leptokurtic standardized residuals and in addition justified in section 4.2. df is always larger than two by taking values between 6.417 (4.174) and 6.432 (4.183) on average for day (night) returns with respect to f_1 and f_2.

We analyze the impact of spillovers for each pair of countries according to the introduced DCC MGARCH(1,1) model. The focus lies in changes of feedback trading during day in Germany and night in the UK. Thereby, results in table 4 serve as a benchmark. For day returns, feedback trading still does not exist in Germany for all estimations as well as both fundamental proxies. In addition, the former positive feedback trading only disappears in the UK. Global interconnectedness between both markets induces a decline of trend chasing investors in the UK. Positive and statistically significant feedback trading is present in the US, Japan, and Hong Kong even with respect to day returns in Germany.

As a result, we identify two important findings for day returns – investors' behavior in Germany is not influenced by all other countries according to feedback trading, but day returns from Germany have an impact on chasing investors in the UK. The picture is partly different when considering night returns. The presence of positive feedback trading is confirmed in Germany, the US, Japan, and Hong Kong at least at the 5% level. More interesting is the interaction between night returns of Germany and the UK. We find a one-sided spillover effect from Germany to the UK when comparing with the benchmark model. The parameter coefficient b_2 is significant positive with a value of 0.010^{*} for the UK with respect to f_1 and f_2 in table 6. This also means that rather negative feedback trading tends to occur in this particular country when including interactions with Germany. Hypothesis 4 can be partly rejected because of cross-country spillover effects as illustrated by the negative feedback trading behavior.[27]

The coefficients for B_t are in line with our previous estimations by taking mostly significant and positive values for night returns. Only a negative parameter coefficient of -0.047^{*} is determined for night returns in the UK with respect to f_2 and an interaction with Germany. Apart from that, the parameter coefficients for day returns are always positive except for Hong Kong as already shown in the feedback trading model.

Not surprisingly, the parameter coefficient b_4 is mostly insignificant for night returns according to f_2 and same reasoning can be applied as before. Significantly negative values of -0.021^{**} and -0.012^{***} are only reported for Japan for f_1

[27] Unfortunately, we cannot give any indication to what extent domestic trading behavior changes. Domestic traders in the UK can still engage equally in positive and negative feedback trading, adhere more to negative feedback trading, or pursue positive trading patterns. However, the total effect discloses a significant amount of negative feedback trading behavior.

Electronic copy available at: https://ssrn.com/abstract=3275317

Table 5: Day returns with global interconnectedness

Coefficient	Germany	UK	Germany	US	Germany	Japan	Germany	Hong Kong
(A) Day returns with f_1 as the fundamental proxy								
b_0	0.091**	0.031	0.045	0.057**	0.078*	0.037	0.069	-0.027
	(0.036)	(0.024)	(0.038)	(0.024)	(0.043)	(0.029)	(0.043)	(0.032)
b_1	-0.062	-0.070	-0.222***	-0.051	-0.282***	-0.292***	-0.298***	0.020
	(0.058)	(0.060)	(0.063)	(0.054)	(0.070)	(0.090)	(0.070)	(0.091)
b_2	0.005	-0.002	-0.001	-0.022***	-0.002	-0.017**	0.001	-0.015**
	(0.005)	(0.005)	(0.006)	(0.005)	(0.007)	(0.008)	(0.007)	(0.007)
b_3	0.112***	0.161***	0.192***	0.092*	0.295***	0.342***	0.292***	0.038
	(0.023)	(0.023)	(0.026)	(0.055)	(0.033)	(0.041)	(0.033)	(0.038)
b_4	0.003	-0.012	0.000**	0.000	0.000*	-0.017**	0.000**	0.000
	(0.018)	(0.017)	(0.020)	(0.000)	(0.022)	(0.008)	(0.043)	(0.000)
α_0	0.023***	0.026***	0.026***	0.022***	0.017**	0.037***	0.179***	0.011**
	(0.007)	(0.007)	(0.008)	(0.006)	(0.007)	(0.009)	(0.007)	(0.004)
α_1	0.075***	0.094***	0.087***	0.138***	0.079***	0.131***	0.081***	0.057***
	(0.011)	(0.013)	(0.014)	(0.019)	(0.015)	(0.020)	(0.015)	(0.011)
β_1	0.908***	0.886***	0.899***	0.853***	0.912***	0.834***	0.908***	0.934***
	(0.013)	(0.016)	(0.016)	(0.018)	(0.016)	(0.024)	(0.159)	(0.129)
ρ	0.674		0.573		0.036		0.108	
λ_1	0.016		0.011		0.002		0.000	
λ_2	0.978		0.981		0.921		0.896	
df	6.709		5.434		6.326		7.197	
Log-likelihood	-5589.084		-5651.589		-5960.890		-6098.745	
(B) Day returns with f_2 as the fundamental proxy								
b_0	0.084***	-0.005	0.105***	0.028	0.114***	0.023	0.117***	-0.041
	(0.029)	(0.026)	(0.031)	(0.024)	(0.035)	(0.029)	(0.035)	(0.272)
b_1	-0.058**	0.041	-0.096***	0.088***	-0.139***	-0.055	-0.141***	-0.005
	(0.025)	(0.026)	(0.028)	(0.028)	(0.034)	(0.040)	(0.033)	(0.033)
b_2	0.005	-0.003	0.000	-0.022***	-0.001	-0.017**	0.002	-0.015**
	(0.005)	(0.004)	(0.006)	(0.005)	(0.007)	(0.008)	(0.007)	(0.006)
b_3	0.107***	0.165***	0.186***	0.109**	0.280***	0.351***	0.277***	0.048
	(0.023)	(0.022)	(0.026)	(0.054)	(0.033)	(0.040)	(0.033)	(0.038)
b_4	-0.018	0.000	-0.004	0.000***	0.000	-0.011***	-0.022	0.000
	(0.013)	(0.011)	(0.015)	(0.000)	(0.019)	(0.004)	(0.019)	(0.000)
α_0	0.023***	0.025***	0.025***	0.022***	0.016**	0.037***	0.018***	0.107**
	(0.007)	(0.007)	(0.008)	(0.006)	(0.007)	(0.009)	(0.007)	(0.004)
α_1	0.073***	0.091***	0.086***	0.138***	0.079***	0.131***	0.080***	0.057***
	(0.011)	(0.013)	(0.014)	(0.019)	(0.015)	(0.020)	(0.015)	(0.011)
β_1	0.909***	0.890***	0.890***	0.853***	0.913***	0.834***	0.909***	0.934***
	(0.013)	(0.015)	(0.015)	(0.018)	(0.016)	(0.024)	(0.016)	(0.013)
ρ	0.674		0.574		0.035		0.109	
λ_1	0.016		0.011		0.010		0.015	
λ_2	0.978		0.981		0.296		0.002	
df	6.714		5.426		6.378		7.211	
Log-likelihood	-5580.503		-5648.747		-5960.044		-6099.705	

Note: This table presents parameter coefficients under the DCC MGARCH(1,1) model. Parameters are estimated simultaneously between day returns of Germany and the UK (the US, Japan as well as Hong Kong) with respect to different fundamental proxies (f_1 and f_2) according to eq. (18) and (19). Stars indicate significance as follows: (*) 10%, (**) 5%, (***) 1%. In parentheses: standard errors.

Electronic copy available at: https://ssrn.com/abstract=3275317

Table 6: Night returns with global interconnectedness

Coefficient	UK	Germany	UK	Japan	UK	Hong Kong
(A) Night returns with f_1 as the fundamental proxy						
b_0	-0.030	-0.008	-0.003	-0.058	-0.017	0.050**
	(0.022)	(0.030)	(0.024)	(0.052)	(0.024)	(0.023)
b_1	-0.156**	-0.122**	-0.208***	-0.105	-0.205***	-0.260***
	(0.065)	(0.049)	(0.076)	(0.105)	(0.076)	(0.063)
b_2	0.010*	-0.009**	0.006	-0.178***	0.003	-0.008**
	(0.005)	(0.004)	(0.006)	(0.016)	(0.006)	(0.004)
b_3	-0.040	0.063***	0.087***	0.099**	0.036	0.059**
	(0.025)	(0.022)	(0.031)	(0.048)	(0.030)	(0.026)
b_4	0.000***	0.000***	0.000**	-0.021**	0.000**	0.000***
	(0.023)	(0.015)	(0.000)	(0.009)	(0.000)	(0.000)
α_0	0.008***	0.025***	0.004***	0.022**	0.005***	0.010***
	(0.002)	(0.007)	(0.001)	(0.010)	(0.002)	(0.004)
α_1	0.029***	0.049***	0.018***	0.042***	0.022***	0.059***
	(0.006)	(0.009)	(0.004)	(0.012)	(0.005)	(0.011)
β_1	0.955***	0.933***	0.971***	0.943***	0.967***	0.940***
	(0.008)	(0.010)	(0.005)	(0.017)	(0.007)	(0.011)
ρ	0.512		0.204		0.154	
λ_1	0.029		0.027		0.013	
λ_2	0.928		0.894		0.961	
df	4.080		4.532		3.911	
Log-likelihood	-4878.089		-5121.652		-5237.738	
(B) Night returns with f_2 as the fundamental proxy						
b_0	0.010	0.038	0.030	-0.059	0.022	0.093***
	(0.017)	(0.029)	(0.018)	(0.052)	(0.018)	(0.021)
b_1	0.003	-0.001	-0.057**	0.161**	-0.028	0.013
	(0.025)	(0.024)	(0.028)	(0.064)	(0.027)	(0.018)
b_2	0.010*	-0.010***	0.006	-0.178***	0.003	-0.008**
	(0.005)	(0.004)	(0.006)	(0.016)	(0.006)	(0.004)
b_3	-0.047*	0.055***	0.082***	0.107**	0.028	0.042
	(0.025)	(0.021)	(0.031)	(0.048)	(0.030)	(0.026)
b_4	0.011	0.001	-0.002	-0.012***	0.007	0.000
	(0.009)	(0.015)	(0.011)	(0.005)	(0.011)	(0.000)
α_0	0.008***	0.026***	0.004***	0.027***	0.006***	0.010***
	(0.002)	(0.007)	(0.001)	(0.012)	(0.002)	(0.004)
α_1	0.029***	0.049***	0.018***	0.045***	0.023***	0.062***
	(0.006)	(0.009)	(0.004)	(0.013)	(0.005)	(0.012)
β_1	0.954***	0.931***	0.971***	0.936***	0.966***	0.937***
	(0.008)	(0.011)	(0.006)	(0.020)	(0.007)	(0.011)
ρ	0.514		0.199		0.158	
λ_1	0.030		0.025		0.014	
λ_2	0.927		0.901		0.959	
df	4.098		4.518		3.934	
Log-likelihood	-4882.328		-5122.664		-5247.037	

Note: This table presents parameter coefficients under the DCC MGARCH(1,1) model. Parameters are estimated simultaneously between night returns of the UK and Germany (Japan as well as Hong Kong) with respect to different fundamental proxies (f_1 and f_2) according to eq. (18) and (19). Stars indicate significance as follows: (*) 10%, (**) 5%, (***) 1%. In parentheses: standard errors.

Electronic copy available at: https://ssrn.com/abstract=3275317

and f_2, respectively, when including the UK and Japan in the bivariate model. The parameter coefficient is less often insignificant for day returns and again only negative for Japan for the interaction between Japan and Germany. Consequently, fewer fundamental traders exist on average in general under global interconnectedness. A possible explanation for this outcome is that stock prices provide less information for fundamental traders. Hinterleitner et al. (2015) argue that a rising uncertainty about fundamental values overnight lowers market quality in terms of liquidity and mispricing. In addition, investors may rely on other fundamentals beside earnings and dividend announcements when considering a multi country setting.

5.3 Robustness and Outlook

We consider a modified demand function for feedback traders as well as different calculation methods for the deviation between the fundamental value and the price in order to verify a sufficient robustness check. With regard to the second verification, the impact of fundamental traders does not change when using a cumulative or weighted moving average of prices instead of actual prices. Based on this, the demand of fundamental traders is rather affected by the underlying fundamental proxy than by the calculation of prices. Koutmos (2012) uses two-year moving averages and the "Fed model"[28] as an additional proxy. He confirms that this proxy is only reliable for the US market. Based on the modification, we consider a demand function of feedback traders that is affected by sentiment in a multiplicative way:

$$Y_{2,t} = [\gamma B_t + \beta(1 - B_t)](R_{t-1} - r_f) \qquad \forall \gamma > 0. \tag{21}$$

Here, the existence of chasing investors depends on the sentiment state and varies for price changes if $\gamma \neq \beta$. Relying on the feedback trading model, we could not find tangible changes for day and night returns with respect to trading behavior. Therefore, we conclude that our suggested eq. (2) is sufficient as the demand function for feedback traders.

Likelihood ratio (LR) tests are calculated to reinforce the relative explanation power of an extended model. LR$= -2 * (L_0 - L_1)$ follows a chi-squared distribution[29], where L_0 is the value of the likelihood function under the null hypothesis and L_1 the value of the alternative hypothesis. Koutmos (2012) demonstrates

[28] The model calculates the yield spread between long-term government bonds and market earnings.

[29] The degrees of freedom depend on the number of parameter restrictions required under the null hypothesis.

Electronic copy available at: https://ssrn.com/abstract=3275317

that an extended model provides advantages in matters of explaining the impact of stock price movements. More related to our global feedback trading model, all log-likelihood values are larger in absolute terms due to a bivariate model with B_t as an additional explanatory dummy variable. Here, we consider the GARCH model as the nested model and test the significance to alternative GARCH specifications. The EGARCH model yields improved log-likelihood values and performs best for most countries in order to calculate the conditional variance.[30]

Our findings are important in understanding investors' trading behavior and what types of indicators they use during different trading hours. The knowledge about the existence of positive or negative feedback trading is beneficial for traders in any given market. A more rational investor could use the diverse implications arising from day and night returns for own arbitrage opportunities by catching incipient trends in stock prices. Investors with low marginal trading costs can develop profitable conditional strategies and time their buy and sell orders better. Thus, strategies have to be adjusted with respect to trading hours and the stock market of interest. An interesting aspect concerning strategies arises by considering intraday data and other asset classes. Market participants diversify their portfolio allocation and do not only invest in stocks. Therefore, a mixed combination with adjusted demand functions has to be developed. As indicated in section 4.1, more renowned sentiment measures can reveal a closer relationship between fundamental and feedback traders in addition. This interaction clarifies mainly why strategies differ across time and countries. All extensions suggest a future research involving behavioral preferences. A detailed discussion of this complex process, however, would exceed the scope of our paper.

6 Conclusion

Feedback trading strategies have gained much popularity among researchers in the last decades and are used to illustrate how new information based on returns is reflected in the markets. Positive feedback traders buy (sell) stocks in a rising (falling) market while negative feedback traders adhere to a contrarian strategy. Stock return autocorrelation increases in the presence of feedback traders. The more feedback traders are active the higher is the autocorrelation due to a greater effect on prices.

The feedback trading model, as historically proposed by Sentana and Wadhwani

[30] All log-likelihood values and likelihood ratio statistics are presented in tables B1 and B2 on page 44f. in the appendix.

Electronic copy available at: https://ssrn.com/abstract=3275317

(1992), is based on the existence of two types of investors. In order to be accurate while using their model, we consider three heterogeneous groups and develop a dynamic feedback trading model by applying a DCC MGARCH model. Thereby, we offer an informative framework for exploring feedback trading from two new angles. First, we decompose the overall return premium into day and night returns. Second, we assume interconnectedness between multiple countries and introduce a global feedback trading model. Dynamic effects with spillovers shed light on investors' strategy and provide an alternative framework about price fluctuations.

Overall, findings confirm the presence of positive feedback traders in daily stock market prices. However, feedback trading strategies differ across markets when distinguishing between day and night returns. Positive feedback trading tends to occur more often for day than night returns due to a decline in trading volume overnight. Market closures affect the mean and standard deviation of returns because investors need to be compensated for the risk when trading overnight. The limited possibility of liquidation constricts night returns to be higher than day returns. A decline in the trading volume during the night diminishes the attractiveness of feedback behavior and the amount of chasing investors is reduced. Feedback traders behave similarly in Japan and Hong Kong during a full trading day, whereas Germany and the UK register a strong contrariness. No feedback trading is found to exist in Germany during trading hours. Simultaneously, the parameter coefficient b_2 is significantly negative for night returns under f_1 as the fundamental proxy. An inverse relationship is found in the UK. During the day, investors follow a positive feedback trading strategy. Missing feedback trading during the night reflects a change in investors' behavior in this particular country. This evidence supports our hypothesis that feedback trading strategies differ during the day and night given a specific market.

Relying on the global feedback trading model, evidence for changing trading behavior is provided by examining the interaction of specific markets. For day returns, feedback trading still does not exist in Germany for all estimations as well as both fundamental proxies. In addition, the former positive feedback trading only disappears in the UK. Global interconnectedness between both markets induces a decline of trend chasing investors in the UK. Positive and statistically significant feedback trading is present in the US, Japan, and Hong Kong even with respect to day returns in Germany. As a result, we identify two important findings for day returns – investors' behavior in Germany is not influenced by all other countries according to feedback trading, but day returns from Germany have an impact on trend chasing investors in the UK. The picture is partly dif-

Electronic copy available at: https://ssrn.com/abstract=3275317

ferent when considering night returns. We find one-sided spillover effects from Germany to the UK when comparing with the benchmark model. The parameter coefficient b_2 is significantly positive for the UK with respect to f_1 and f_2. This also means that it is negative feedback trading that tends to occur in this particular country when including interactions with Germany.

All in all, feedback trading strategies differ during day and night with spillover effects arising from market interactions. Future research should include a broader range of different asset classes and more renowned sentiment measures to understand behavioral preferences as well as intraday data to capture spillover effects during contemporaneous stock exchange openings.

Electronic copy available at: https://ssrn.com/abstract=3275317

References

Aielli, G. P. (2009). Dynamic Conditional Correlations: On Properties and Estimation. *Working paper*, Dipartimento di Statistica, University of Florence, Florence, Italy.

Antoniou, A. and G. Koutmos (2014). The Cost of Credit and Positive Feedback Trading: Evidence from the UK Stock Market. *Journal of Applied Finance and Banking 4*(2), 21–32.

Antoniou, A., G. Koutmos, and A. Pericli (2005). Index futures and positive feedback trading: evidence from major stock exchanges. *Journal of Empirical Finance 12*(2), 219–238.

Aretz, K. and S. M. Bartram (2015). Making Money While You Sleep? Anomalies in International Day and Night Returns. *Working paper*.

Baker, M. and J. Wurgler (2006). Investor sentiment and the cross-section of stock returns. *The Journal of Finance 61*(4), 1645–1680.

Baker, M. and J. Wurgler (2007). Investor Sentiment in the Stock Market. *Journal of Economic Perspectives 21*(2), 129–151.

Barclay, M. J. and T. Hendershott (2003). Price Discovery and Trading After Hours. *Review of Financial Studies 16*(4), 1041–1073.

Bauwens, L., S. Laurent, and J. V. K. Rombouts (2006). Multivariate GARCH models: a survey. *Journal of Applied Econometrics 21*(1), 79–109.

Becker, K. G., J. E. Finnerty, and J. Friedman (1995). Economic news and equity market linkages between the U.S. and U.K. *Journal of Banking & Finance 19*(7), 1191–1210.

Beja, A. and M. B. Goldman (1980). On The Dynamic Behavior of Prices in Disequilibrium. *The Journal of Finance 35*(2), 235–248.

Berndt, E. R., B. H. Hall, R. E. Hall, and J. A. Hausman (1974). Estimation and inference in nonlinear structural models. *Annals of Economic and Social Measurement 3*(4), 653–665.

Blanc, P., R. Chicheportiche, and J.-P. Bouchaud (2014). The fine structure of volatility feedback II: Overnight and intra-day effects. *Physica A: Statistical Mechanics and its Applications 402*, 58–75.

Electronic copy available at: https://ssrn.com/abstract=3275317

Bohl, M. T., A. C. Klein, and P. L. Siklos (2013). Are short sellers positive feedback traders? Evidence from the global financial crisis. *Journal of Financial Stability 9*(3), 337–346.

Bohl, M. T. and P. L. Siklos (2008). Empirical evidence on feedback trading in mature and emerging stock markets. *Applied Financial Economics 18*(17), 1379–1389.

Bollerslev, T., R. F. Engle, and J. M. Wooldridge (1988). A capital asset pricing model with time-varying covariances. *Journal of political Economy 96*(1), 116–131.

Bollerslev, T. and J. M. Wooldridge (1992). Quasi-maximum likelihood estimation and inference in dynamic models with time-varying covariances. *Econometric Reviews 11*(2), 143–172.

Campbell, J. Y. and L. Hentschel (1992). No news is good news: An asymmetric model of changing volatility in stock returns. *Journal of Financial Economics 31*(3), 281–318.

Campbell, J. Y. and A. S. Kyle (1993). Smart Money, Noise Trading and Stock Price Behaviour. *The Review of Economic Studies 60*(1), 1–34.

Caporin, M. and M. McAleer (2012). Do We Really Need Both BEKK And DCC? A Tale Of Two Multivariate GARCH Models. *Journal of Economic Surveys 26*(4), 736–751.

Charteris, A., F. Chau, K. Gavriilidis, and V. Kallinterakis (2014). Premiums, discounts and feedback trading: Evidence from emerging markets' ETFs. *International Review of Financial Analysis 35*, 80–89.

Chau, F., R. Deesomsak, and M. C. Lau (2011). Investor sentiment and feedback trading: Evidence from the exchange-traded fund markets. *International Review of Financial Analysis 20*(5), 292–305.

Chau, F., J.-M. Kuo, and Y. Shi (2015). Arbitrage opportunities and feedback trading in emissions and energy markets. *Journal of International Financial Markets, Institutions and Money 36*, 130–147.

Conrad, J., G. Kaul, and M. Nimalendran (1991). Components of short-horizon individual security returns. *Journal of Financial Economics 29*(2), 365–384.

Cooper, M., M. Cliff, and H. Gulen (2008). Return differences between trading and non-trading hours: Like night and day. *Working paper*.

Electronic copy available at: https://ssrn.com/abstract=3275317

Cutler, D. M., J. M. Poterba, and L. H. Summers (1990). Speculative dynamics and the role of feedback traders. *American Economic Review 80*(2), 63–68.

De Long, J. B., A. Shleifer, L. H. Summers, and R. J. Waldmann (1990). Noise trader risk in financial markets. *Journal of Political Economy 98*(4), 703–738.

Dean, W. G. and R. W. Faff (2008). Evidence of feedback trading with Markov switching regimes. *Review of Quantitative Finance and Accounting 30*(2), 133–151.

Engle, R. (2002). Dynamic conditional correlation: A simple class of multivariate generalized autoregressive conditional heteroskedasticity models. *Journal of Business & Economic Statistics 20*(3), 339–350.

Fama, E. F. (1970). Efficient Capital Markets: A Review of Theory and Empirical Work. *The Journal of Finance 25*(2), 383–417.

Fama, E. F. and K. R. French (2002). The equity premium. *The Journal of Finance 57*(2), 637–659.

Farmer, J. D. and S. Joshi (2002). The price dynamics of common trading stratgies. *Journal of Economic Behavior & Organization 49*(2), 149–171.

Francis, J., D. Pagach, and J. Stephan (1992). The Stock Market Response to Earnings Announcements Released During Trading versus Nontrading Periods. *Journal of Accounting Research 30*(2), 165.

Frino, A. and A. Hill (2000). Intranight trading behaviour. *Working paper*.

Giannetti, A., S. J. Larson, C. I. Lee, and J. Madura (2006). Price movements, information, and liquidity in the night trading market. *Financial Review 41*(1), 119–137.

Gordon, M. J. (1959). Dividends, Earnings, and Stock Prices. *The Review of Economics and Statistics 41*(2), 99.

Greene, J. T. and S. G. Watts (1996). Price Discovery on the NYSE and the NASDAQ: The Case of Overnight and Daytime News Releases. *Financial Management 25*(1), 19.

Hagerman, R. L. (1978). More Evidence on the Distribution of Security Returns. *The Journal of Finance 33*(4), 1213.

Hinterleitner, G., U. Leopold-Wildburger, R. Mestel, and S. Palan (2015). A Good Beginning Makes a Good Market: The Effect of Different Market Opening Structures on Market Quality. *The Scientific World Journal 2015*, 1–14.

Electronic copy available at: https://ssrn.com/abstract=3275317

Hong, H. and J. Wang (2000). Trading and returns under periodic market closures. *The Journal of Finance 55*(1), 297–354.

Hsieh, M.-F., T.-Y. Yang, Y.-T. Yang, and J.-S. Lee (2011). Evidence of herding and positive feedback trading for mutual funds in emerging Asian countries. *Quantitative Finance 11*(3), 423–435.

Jain, R. (2007). Institutional and individual investor preferences for dividends and share repurchases. *Journal of Economics and Business 59*(5), 406–429.

Jegadeesh, N. and S. Titman (1993). Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency. *The Journal of Finance 48*(1), 65–91.

Koulakiotis, A. and A. Kiohos (2016). Positive feedback trading and long-term volatility links: evidence from real estate markets of USA, Be/Lux and Switzerland. *Applied Economics Letters 23*(2), 97–100.

Koutmos, D. (2012). An intertemporal capital asset pricing model with heterogeneous expectations. *Journal of International Financial Markets, Institutions and Money 22*(5), 1176–1187.

Koutmos, G. (1997). Feedback trading and the autocorrelation pattern of stock returns: further empirical evidence. *Journal of International Money and Finance 16*(4), 625–636.

Koutmos, G. (2014). Positive feedback trading: a review. *Review of Behavioural Finance 6*(2), 155–162.

Kurov, A. (2008). Investor sentiment, trading behavior and informational efficiency in index futures markets. *Financial Review 43*(1), 107–127.

Kwon, K.-Y. and R. J. Kish (2002). Technical trading strategies and return predictability: NYSE. *Applied Financial Economics 12*(9), 639–653.

Lakonishok, J., A. Shleifer, and R. W. Vishny (1992). The impact of institutional trading on stock prices. *Journal of Financial Economics 32*(1), 811–827.

Lamont, O. (1998). Earnings and expected returns. *The Journal of Finance 53*(5), 1563–1587.

LeBaron, B. (1992). Some relations between volatility and serial correlations in stock market returns. *Journal of Business 65*(2), 199–219.

Lee, H.-J. (2016). Individuals' feedback trading in market and limit trades: Trading behaviours on the Korean stock market. *Investment Analysts Journal 45*(3), 212–232.

Electronic copy available at: https://ssrn.com/abstract=3275317

Lo, A. W. and A. C. MacKinlay (1990). When are contrarian profits due to stock market overreaction? *Review of Financial Studies 3*(2), 175–205.

Long, J. B. D., A. Shleifer, L. H. Summers, and R. J. Waldmann (1989). The Size and Incidence of the Losses from Noise Trading. *The Journal of Finance 44*(3), 681–696.

Longin, F. (2005). The choice of the distribution of asset returns: How extreme value theory can help? *Journal of Banking & Finance 29*(4), 1017–1035.

Longstaff, F. A. (1995). How Much Can Marketability Affect Security Values? *The Journal of Finance 50*(5), 1767.

Lou, D., C. Polk, and S. Skouras (2015). A tug of war: Overnight versus intraday expected returns. *Working paper*.

Madhavan, A., M. Richardson, and M. Roomans (1997). Why do security prices change? A transaction-level analysis of NYSE stocks. *The Review of Financial Studies 10*(4), 1035–1064.

Merton, R. C. (1973). An Intertemporal Capital Asset Pricing Model. *Econometrica 41*(5), 867–887.

Nelson, D. B. (1991). Conditional Heteroskedasticity in Asset Returns: A New Approach. *Econometrica 59*(2), 347–370.

Officer, R. R. (1972). The Distribution of Stock Returns. *Journal of the American Statistical Association 67*(340), 807.

Oldfield, G. S. and R. J. Rogalski (1979). Stock Returns Over Open and Closed Trading Periods. *The Journal of Financial and Quantitative Analysis 14*(4), 807.

Patell, J. M. and M. A. Wolfson (1982). Good news, bad news, and the intraday timing of corporate disclosures. *Accounting Review*, 509–527.

Patell, J. M. and M. A. Wolfson (1984). The intraday speed of adjustment of stock prices to earnings and dividend announcements. *Journal of Financial Economics 13*(2), 223–252.

Peiró, A. (1994). The distribution of stock returns: international evidence. *Applied Financial Economics 4*(6), 431–439.

Sentana, E. and S. Wadhwani (1992). Feedback Traders and Stock Return Autocorrelations: Evidence from a Century of Daily Data. *The Economic Journal 102*(411), 415–425.

Electronic copy available at: https://ssrn.com/abstract=3275317

Shefrin, H. and M. Statman (1985). The Disposition to Sell Winners Too Early and Ride Losers Too Long: Theory and Evidence. *The Journal of Finance 40*(3), 777–790.

Shiller, R. J. (1990). Market Volatility and Investor Behavior. *American Economic Association, Papers and Proceedings 80*(2), 58–62.

Shiller, R. J., S. Fischer, and B. M. Friedman (1984). Stock Prices and Social Dynamics. *Brookings Papers on Economic Activity 2*, 457–510.

Silvennoinen, A. and T. Teräsvirta (2009). Multivariate GARCH models. In *Handbook of financial time series*, pp. 201–229. Springer.

Spiegel, M. and A. Subrahmanyam (1995). On Intraday Risk Premia. *The Journal of Finance 50*(1), 319.

Stoll, H. R. and R. E. Whaley (1990). Stock Market Structure and Volatility. *The Review of Financial Studies 3*(1), 37–71.

Tavakkol, A. (2000). Positive feedback trading in the options market. *Quarterly Journal of Business and Economics*, 69–80.

Watanabe, T. (2002). Margin requirements, positive feedback trading, and stock return autocorrelations: the case of Japan. *Applied Financial Economics 12*(6), 395–403.

White, H. (1982). Maximum Likelihood Estimation of Misspecified Models. *Econometrica 50*(1), 1–25.

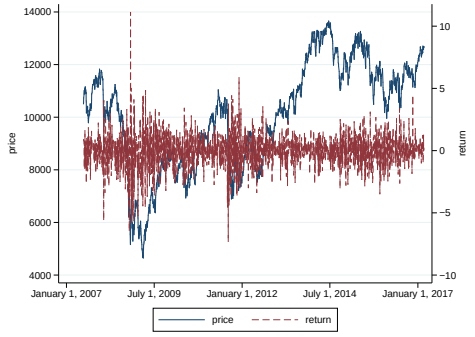
Wood, R. A., T. H. McInish, and J. K. Ord (1985). An Investigation of Transactions Data for NYSE Stocks. *The Journal of Finance 40*(3), 723.

Electronic copy available at: https://ssrn.com/abstract=3275317

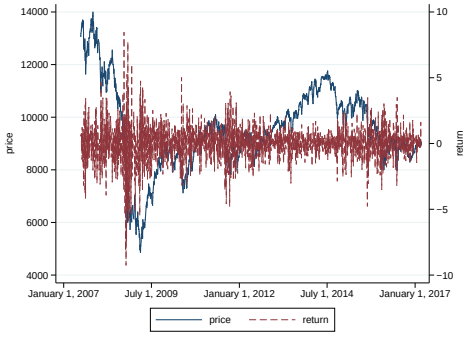
Appendix

A. Figures

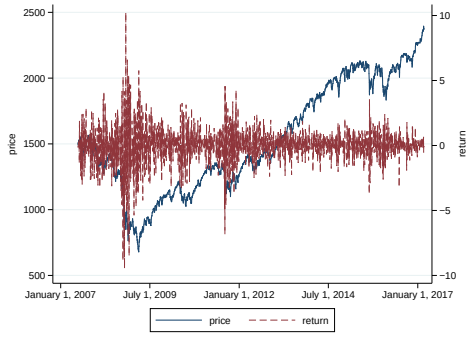
Figure A1: Opening prices and day returns



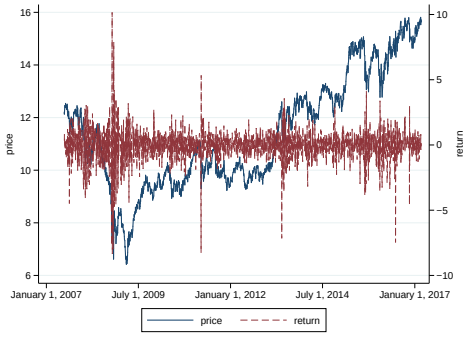
(a) Germany



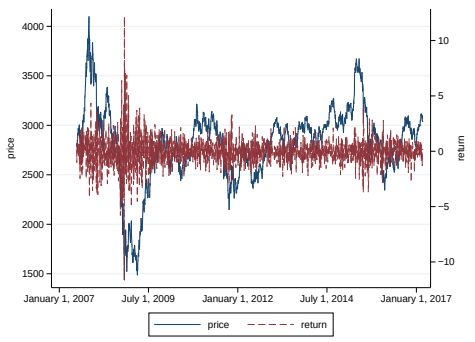
(b) UK



(c) US



(d) Japan

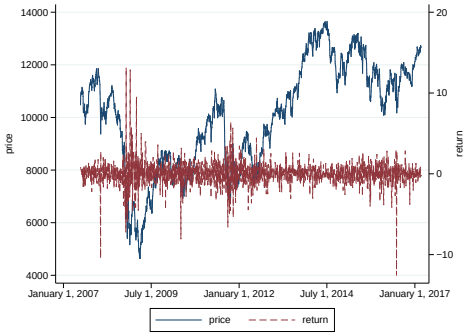


(e) Hong Kong

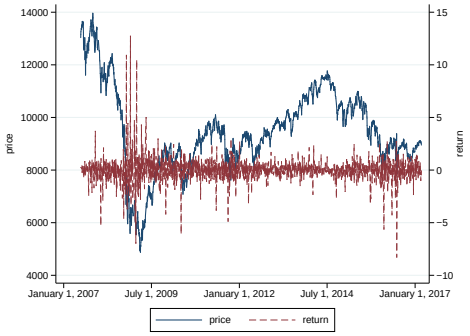
Note: This figure provides opening prices and day returns for each market on a daily basis from June 27, 2007 to March 7, 2017. All returns are in logarithmic first differences and expressed as a percentage. All foreign opening prices are indicated in USD as a common currency basis.

Electronic copy available at: https://ssrn.com/abstract=3275317

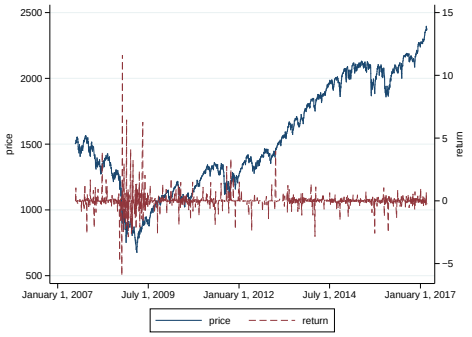
Figure A2: Closing prices and night returns



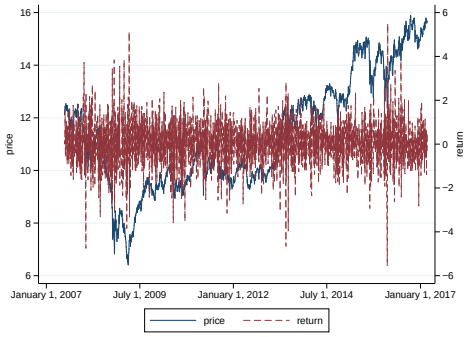
(a) Germany



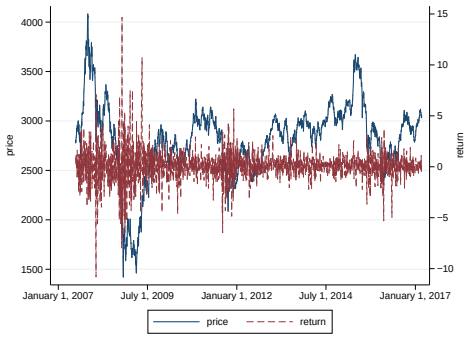
(b) UK



(c) US



(d) Japan

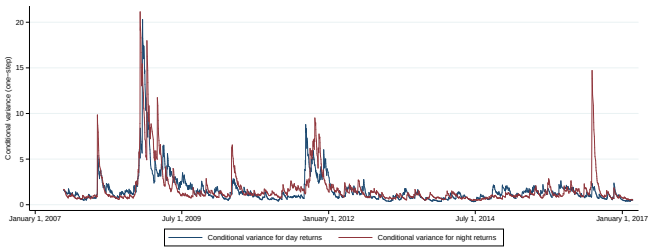


(e) Hong Kong

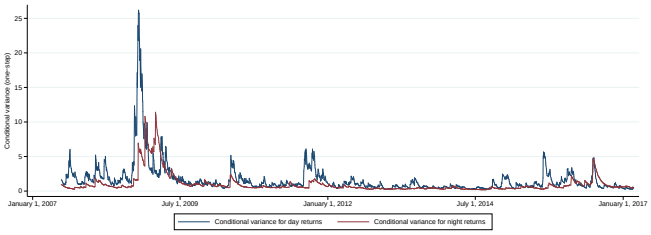
Note: This figure provides closing prices and night returns for each market on a daily basis from June 27, 2007 to March 7, 2017. All returns are in logarithmic first differences and expressed as a percentage. All foreign closing prices are indicated in USD as a common currency basis.

Electronic copy available at: https://ssrn.com/abstract=3275317

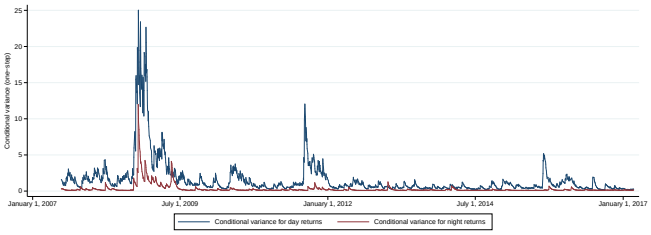
Figure A3: Conditional variances of returns



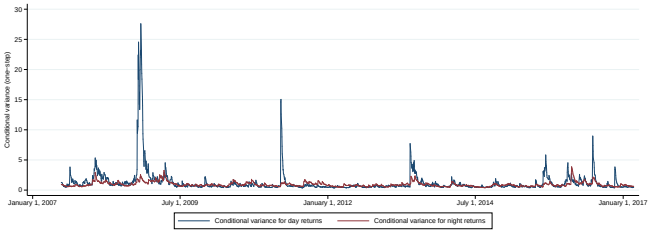
(a) Germany



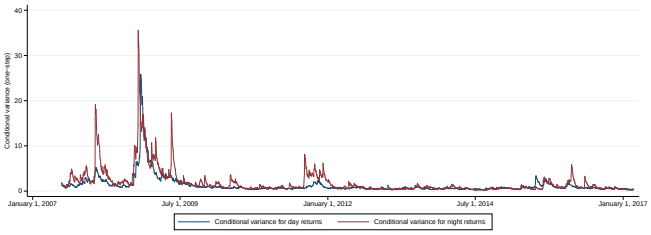
(b) UK



(c) US



(d) Japan

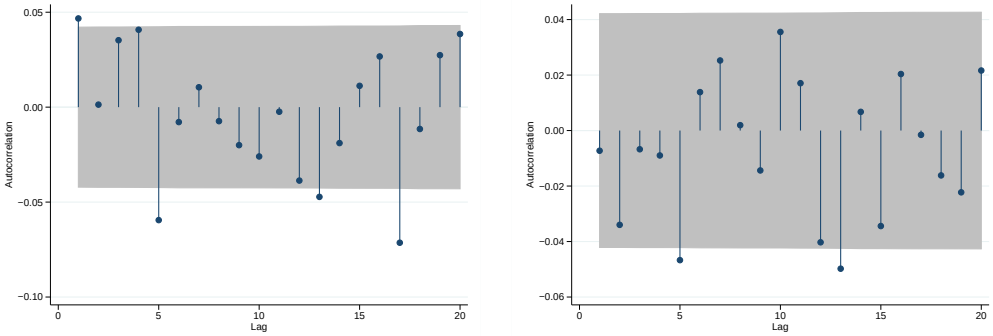


(e) Hong Kong

Note: Conditional variances of day and night returns are presented on a daily basis from June 27, 2007 to March 7, 2017. In general, variances of day returns exhibit larger magnitudes.

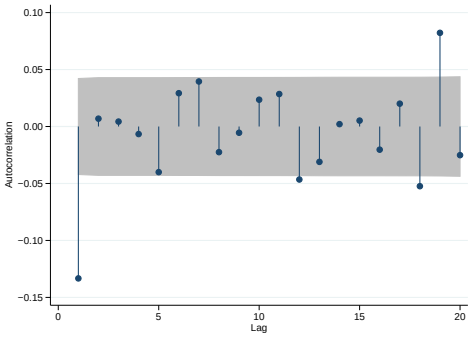
Electronic copy available at: https://ssrn.com/abstract=3275317

Figure A4: Autocorrelations for day returns

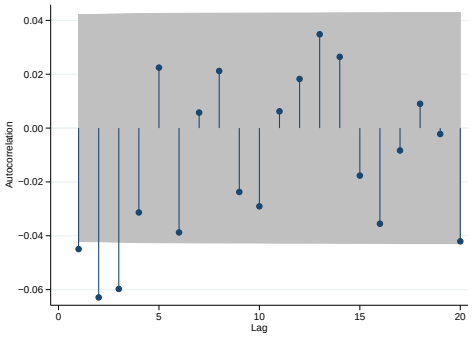


(a) Germany

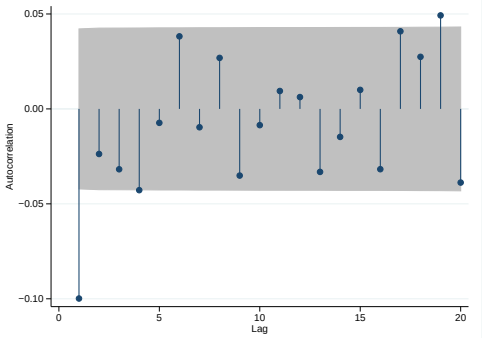
(b) UK



(c) US



(d) Japan

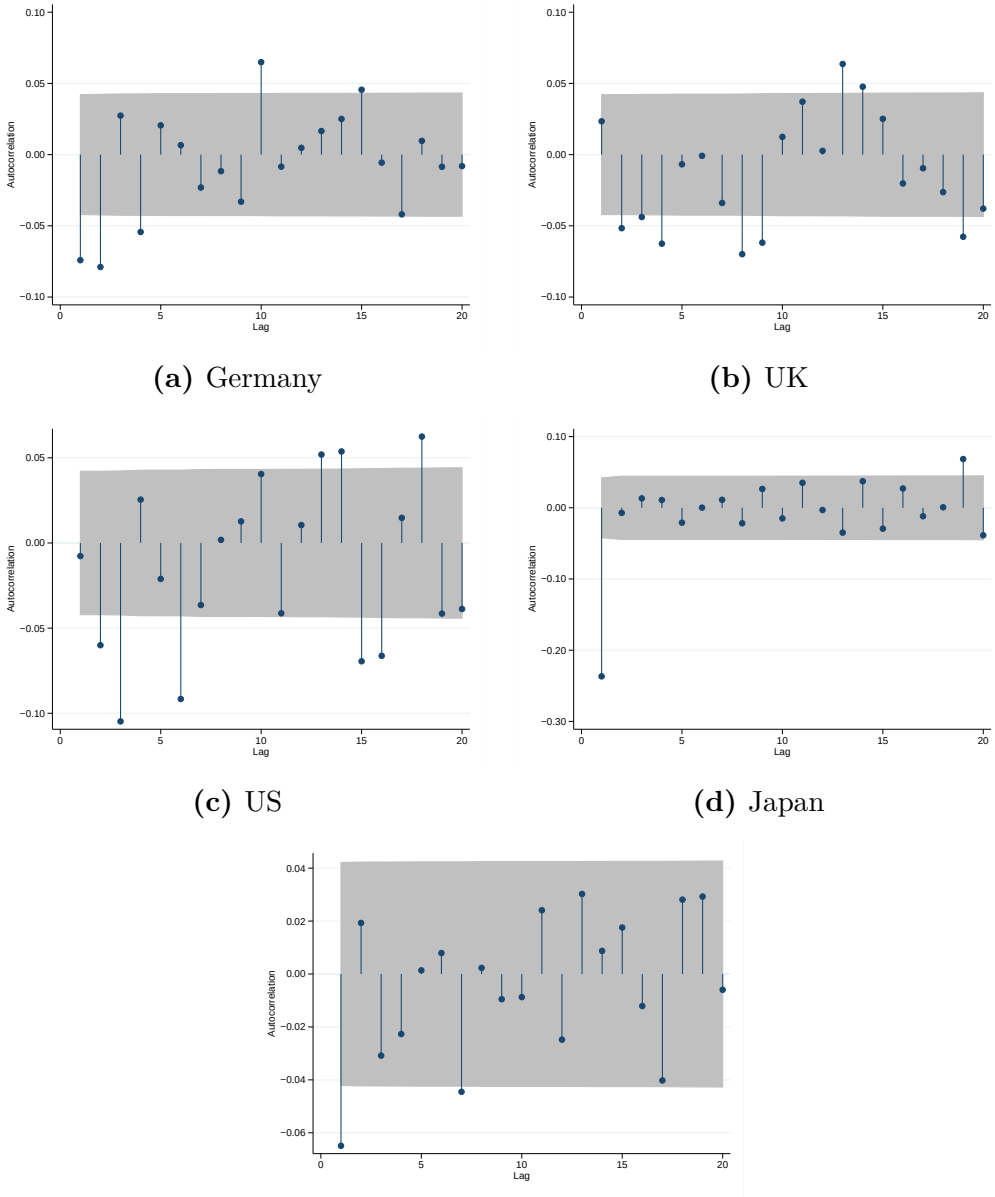


(e) Hong Kong

Note: This figure provides autocorrelations of day returns up to 20 lags for each market. Confidence bands depict the variance at the 95% level according to Bartlett's formula. The EGARCH(1,1) model can be adopted since most autocorrelations are significant different from zero at least with a lag of one.

Electronic copy available at: https://ssrn.com/abstract=3275317

Figure A5: Autocorrelations for night returns



(a) Germany **(b)** UK

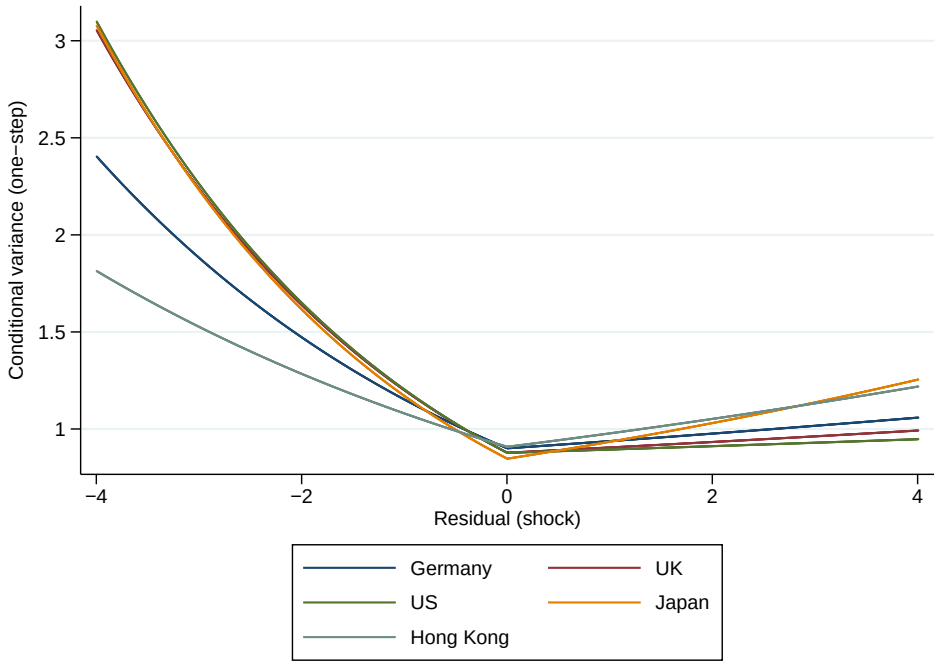
(c) US **(d)** Japan

(e) Hong Kong

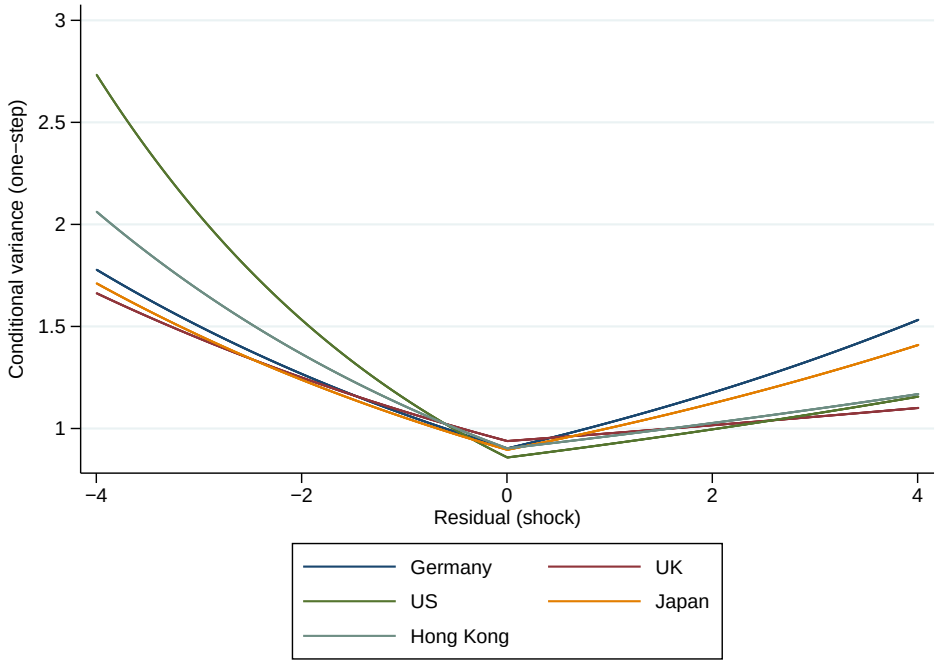
Note: This figure provides autocorrelations of night returns up to 20 lags for each market. Confidence bands depict the variance at the 95% level according to Bartlett's formula. The EGARCH(1,1) model can be adopted since most autocorrelations are significant different from zero at least with a lag of one.

Electronic copy available at: https://ssrn.com/abstract=3275317

Figure A6: News response functions



(a) day returns



(b) night returns

Note: News response functions illustrate the impact of unanticipated shocks on the conditional variance for day and night returns. The conditional variance rises faster for negative residuals than for positive ones according to an asymmetric pattern and larger shocks have a greater impact on volatility.

Electronic copy available at: https://ssrn.com/abstract=3275317

B. Tables

Table B1: Log-likelihood

Coefficient	Germany	UK	US	Japan	Hong Kong
(A) Day returns with f_1 as the fundamental proxy					
GARCH	-3197.634	-3103.333	-2908.677	-2791.915	-2908.871
EGARCH	**-3168.477**	**-3066.954**	**-2878.993**	-2779.925	**-2899.521**
GJR	-3177.782	-3068.274	-2882.988	-2784.635	-2901.666
PARCH	-3193.760	-3102.101	-2908.674	-2791.284	-2906.895
NARCH	-3171.051	-3072.120	-2879.805	**-2777.907**	-2902.946
(B) Day returns with f_2 as the fundamental proxy					
GARCH	-3198.849	-3092.408	-2902.423	-2789.708	-2908.623
EGARCH	**-3168.725**	**-3062.940**	-2880.079	-2780.636	**-2897.983**
GJR	-3178.466	-3064.530	-2879.747	-2783.598	-2899.732
PARCH	-3194.892	-3091.271	-2902.404	-2789.489	-2906.431
NARCH	-3171.592	-3066.276	**-2878.547**	**-2777.383**	-2901.789
(C) Night returns with f_1 as the fundamental proxy					
GARCH	-3137.391	-2325.400		-2902.605	-3081.527
EGARCH	**-3124.226**	**-2313.100**		**-2898.319**	**-3064.398**
GJR	-3136.892	-2325.115		-2902.525	-3070.411
PARCH	-3124.500	-2316.438		-2899.168	-3076.035
NARCH	-3137.311	-2322.890		-2902.071	-3069.067
(D) Night returns with f_2 as the fundamental proxy					
GARCH	-3140.977	-2330.628		-2902.256	-3086.619
EGARCH	**-3127.728**	**-2317.743**		**-2897.535**	**-3068.330**
GJR	-3140.740	-2330.252		-2902.075	-3073.930
PARCH	-3128.634	-2321.855		-2898.400	-3082.091
NARCH	-3140.826	-2327.458		-2901.331	-3073.027

Note: This table reports log-likelihood values according to a selection of GARCH models with respect to day and night returns as well as both fundamental proxies (f_1 and f_2).

Electronic copy available at: https://ssrn.com/abstract=3275317

Table B2: Likelihood ratio test

Coefficient	Germany	UK	US	Japan	Hong Kong
(A) Day returns with f_1 as the fundamental proxy					
EGARCH	58.314***	72.758***	59.368***	23.980***	18.700***
GJR	39.704***	70.118***	51.378***	14.560***	14.410***
PARCH	7.748***	2.464	0.006	1.262	3.952**
NARCH	53.166***	64.426***	57.744***	28.016***	11.850***
(B) Day returns with f_2 as the fundamental proxy					
EGARCH	60.248***	58.936***	44.688***	18.144***	21.280***
GJR	40.766***	55.756***	45.352***	12.220***	17.782***
PARCH	7.914***	2.274	0.038	0.438	4.564**
NARCH	54.514***	52.264***	47.752***	24.650***	13.668***
(C) Night returns with f_1 as the fundamental proxy					
EGARCH	26.330***	24.600***		8.572***	34.258***
GJR	0.998	0.570		0.160	22.232***
PARCH	25.782***	17.924***		6.874***	10.984***
NARCH	0.160	5.020**		1.068	24.920***
(D) Night returns with f_2 as the fundamental proxy					
EGARCH	26.498***	25.770***		9.442***	36.578***
GJR	0.474	0.752		0.362	25.378***
PARCH	24.686***	17.546***		7.712***	9.056***
NARCH	0.302	6.340**		1.850	27.184***

Note: This table illustrates the LR statistic in order to test the significance of the extended models (EGARCH, GJR, PARCH, and NARCH) to the nested model (GARCH) with respect to day and night returns as well as different fundamental proxies (f_1 and f_2). The LR statistic is calculated by LR$= -2 * (L_0 - L_1)$. Stars indicate significance as follows: (*) 10%, (**) 5%, (***) 1%.

Electronic copy available at: https://ssrn.com/abstract=3275317